Inflation Dynamics - Computational Notebook

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Abstract

This computational notebook documents the data construction, model specifications, estimation workflows, and best-model reporting used to study U.S. quarterly inflation dynamics.

This notebook is organized around five pieces.

1. **Data summary.** The data section constructs the effective inflation sample **1969Q2–2022Q4**, with **215 observations** and give statistical summary.
2. **Mean processes.** The mean-process menu includes Constant, ARX(1,1), ARX(2,1), and ARX(2,2) specifications. These define $\hat{\pi}_t$ and therefore the residual sequence passed to volatility models.
3. **GARCH-family volatility models.** The GARCH section estimates GARCH, GJR-GARCH, and EGARCH specifications with normal, standardized Student's t, and Gaussian-mixture residuals.
4. **Regime-switching models.** The regime-switching section studies state-dependent mean dynamics with normal residuals.
5. **BEGE-GARCH models.** The BEGE section estimates Constant, Symmetric, InflationDeflation, BadGood, Constant-p Full BEGE, Constant-n Full BEGE, and Full BEGE models.

The following notation is used throughout:

- π_t: realized inflation.
- $\mathcal{F}_{t-1}$: information set available at time $t-1$.
- SPF_{t-1}: professional forecast of inflation π_t based on $\mathcal{F}_{t-1}$.
- $\hat{\pi}_t$: expected inflation of selected mean process, $E[\pi_t \mid \mathcal{F}_{t-1}]$.
- u_t: inflation innovation, defined as $u_t := \pi_t - \hat{\pi}_t$.
- u_t^+: positive residual component, $u_t^+ := u_t \cdot \mathbf{1}(u_t > 0)$.
- u_t^-: negative residual component, $u_t^- := u_t \cdot \mathbf{1}(u_t \leq 0)$.

Computation Notebook

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1. Inflation Summary Statistics

This section provides a statistical summary of quarterly inflation and the professional forecast over time.

1.a. Data Sources and Construction

The workbook `Aggregate_CPI_inflation_YYYYMMDD.xlsx` (currently `Aggregate_CPI_inflation_20260714.xlsx`; the original snapshot is `Aggregate_CPI_inflation_20230513.xls`) is built from two public sources:

1. **CPIAUCSL** (FRED, St. Louis Fed): Consumer Price Index for All Urban Consumers: All Items in U.S. City Average, Index 1982–1984=100, Monthly, Seasonally Adjusted. https://fred.stlouisfed.org/series/CPIAUCSL
2. **Survey of Professional Forecasters** (Philadelphia Fed): median *level* forecasts of the GNP/GDP price deflator (`Median_PGDP_Level.xlsx`). https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/survey-of-professional-forecasters

Construction rules (verified to reproduce every value in the original 2023 workbook exactly):

- **Quarterly price index**: the CPIAUCSL level in the *last month* of each quarter; quarterly inflation π_t is its percent change.
- **Monthly inflation**: percent change of monthly CPIAUCSL.
- **SPF expected inflation** for quarter t: $(PGDP3/PGDP2 - 1) \times 100$ from the survey taken in quarter $t - 1$, i.e. the median one-quarter-ahead deflator forecast over the median nowcast. The first target quarter is 1969Q1 (from the 1968Q4 survey).
- **Monthly SPF expected inflation**: the quarterly value de-compounded geometrically, $\left((1 + q/100)^{1/3} - 1\right) \times 100$, held constant within the quarter.
- **Inflation shock**: realized inflation minus SPF expected inflation, at both frequencies.

The workbook and the pickled estimation datasets are regenerated by `update_inflation_data.py` (downloads both sources and rewrites the files; pass `--write-quarterly-pkl` to also regenerate the quarterly pickle). On the overlap with the 2023 snapshot, the SPF series is identical at every date; CPIAUCSL levels and inflation differ only from 2019 onward (max 0.29 index points, max 0.17 percentage points of monthly inflation) because BLS revises recent seasonal-adjustment factors each year. Both pickles were regenerated on 2026-07-14: the quarterly effective sample now runs 1969Q2–2026Q1 (228 observations) and the monthly effective sample 1969M2–2026M5 (688 observations). Quarterly estimation results produced before this update (GARCH, regime-switching, BEGE, OLS mean processes) were estimated on the previous 1969Q2–2022Q4 (215-observation, 2023-vintage) sample and remain comparable with each other but not with re-estimates on the updated sample.

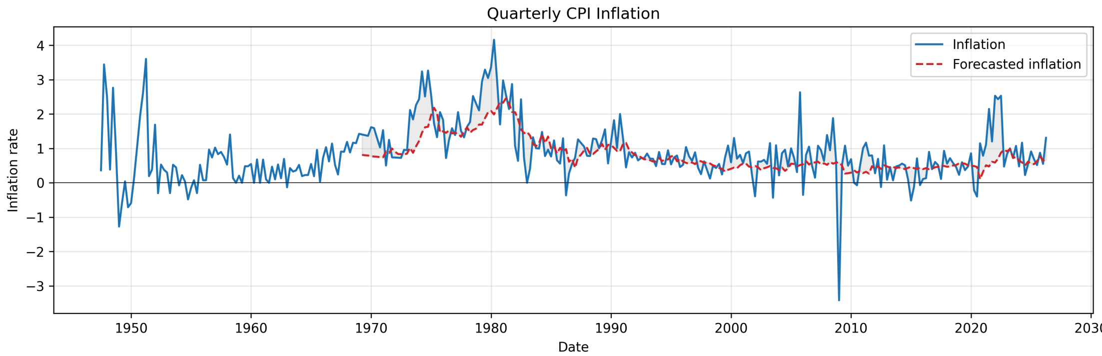


Figure 1: Quarterly inflation π_t and professional forecast SPF_{t-1}.

Inflation data are available from 1947, but the professional forecast series begins in 1969.

	Inflation	Forecasted inflation
Date Start	1947Q2	1969Q1
Date End	2026Q1	2026Q1
#Observations	316	229

Table 1: Quarterly Inflation Statistics

1.b. Effective Sample

To make log-likelihood values comparable across specifications, I use the same effective sample size for all mean models and all volatility models.[1]

In ARX models, lagged regressors (for example, SPF_{t-1}) are not observed at the very beginning of the raw sample. Therefore, I start the estimation sample at the first date where both current forecast inflation (SPF_t), and one-lag forecast inflation (SPF_{t-1}) are available. For the quarterly data, forecast inflation (SPF_t) is available from **1969Q1** to **2026Q1** (229 observations), requiring one lag (SPF_{t-1}) shifts the usable start to **1969Q2**. So the trimmed estimation window is **1969Q2–2026Q1**, with **228 observations**. This same trimmed sample is then used for all mean specifications, so the residual series has the same length across models.

Estimation results committed before the 2026-07-14 data update were produced on the previous trimmed window, **1969Q2–2022Q4** (**215 observations**, 2023 data vintage). Those results remain internally comparable, but their likelihood values are not comparable with estimates on the updated 228-observation sample until the models are re-estimated.

For the reported GARCH-family estimates, conditional-variance recursion starts use the default initialization implemented by the `arch` package. The estimator therefore no longer iterates the pre-sample recursion to force the initial variance to equal a parameter-implied unconditional variance.

1.c. Effective Sample Summary Statistics

[1]Earlier versions of the estimation code contained a sample-trimming mistake and therefore used different numbers of observations across model settings. For example, some GARCH runs used 208 observations, while some BEGE runs used 210 observations. This coding error has now been corrected: all reported and regenerated estimates use the common **1969Q2–2022Q4** effective sample with **215 observations**. As a result, some current log-likelihood, AIC, and BIC values differ from earlier archived numbers.

Summary statistics for the trimmed effective sample are reported below. Skewness and excess kurtosis are bias-corrected for sample size.

	Inflation (π)	SPF	$\pi - SPF$
Date Start	1969Q2	1969Q2	1969Q2
Date End	2026Q1	2026Q1	2026Q1
#Observations	228	228	228
Mean	0.9791	0.8224	0.1567
Median	0.7914	0.6310	0.1052
Std	0.8389	0.4849	0.6499
Min	−3.4170	0.1048	−4.0117
Max	4.1612	2.4566	2.1729
P5	−0.0435	0.3542	−0.7295
P25	0.5216	0.4806	−0.1645
P75	1.2741	0.9874	0.4608
P95	2.5612	1.9801	1.2397
Skewness	0.4033	1.4374	−0.6886
Excess kurtosis	4.1519	1.4855	7.6198
AC(1)	0.5907	0.9706	0.2680
AC(2)	0.5326	0.9458	0.1298
AC(4)	0.4596	0.8892	0.0795
AC(12)	0.2685	0.6897	−0.0015

Table 2: Inflation statistics

1.d. Monthly Data

The monthly counterpart to the quarterly estimation dataset is `Aggregate_CPI_inflation_Monthly.pkl`, built by `update_inflation_data.py` from the updated workbook with the same trimming rule: the sample starts at the first month where both SPF_t and SPF_{t-1} are available. Monthly SPF is available from **1969M1**, so the trimmed monthly estimation window is **1969M2–2026M5**, with **688 observations**. The dataset carries the same columns as the quarterly pickle (`Inflation`, `Inflation_lag_1`, `Inflation_lag_2`, `SPF`, `SPF_lag_1`, `SPF_shock`); note that `SPF` and `SPF_lag_1` are the geometrically de-compounded quarterly forecasts and are therefore constant within each quarter.

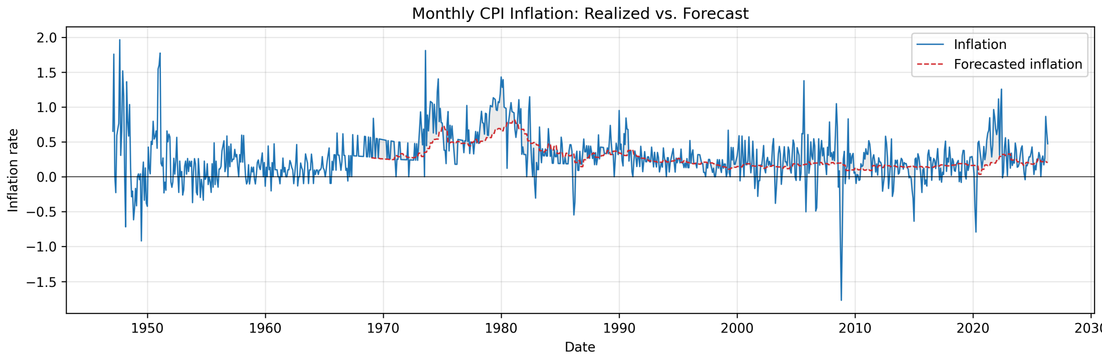


Figure 2: Monthly inflation π_t and professional forecast SPF_{t-1}.

Summary statistics for the trimmed monthly effective sample (generated by `monthly_summary.py`):

	Inflation (π)	SPF	$\pi - SPF$
Date Start	1969-02	1969-02	1969-02
Date End	2026-05	2026-05	2026-05
#Observations	688	688	688
Mean	0.3260	0.2729	0.0531
Median	0.2751	0.2099	0.0415
Std	0.3242	0.1597	0.2736
Min	−1.7705	0.0349	−1.9684
Max	1.8100	0.8123	1.4582
P5	−0.0750	0.1160	−0.3390
P25	0.1648	0.1600	−0.0779
P75	0.4776	0.3271	0.1893
P95	0.9523	0.6584	0.4787
Skewness	0.1761	1.4311	−0.4131
Excess kurtosis	4.1945	1.4523	6.7279
AC(1)	0.6266	0.9902	0.4636
AC(2)	0.4439	0.9804	0.1968
AC(4)	0.4038	0.9623	0.1149
AC(12)	0.2932	0.8894	−0.0080

Table 3: Monthly inflation statistics

2. Mean Process Specification

Four types of mean processes are of interest:

- **Constant**: $\pi_{t+1} = SPF_t + u_{t+1}$
- **ARX(1,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + u_{t+1}$
- **ARX(2,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + u_{t+1}$
- **ARX(2,2)** : $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + u_{t+1}$

where u_{t+1} is the residual term.

2.a. OLS Results

Estimation results of four mean models with HAC standard errors in parentheses below each estimate. (The log-likelihood is computed under the Gaussian assumption.)

Constant:

$$\hat{\pi}_{t+1} = SPF_t \tag{1}$$

No estimated coefficients. Residuals $u_{t+1} = \pi_{t+1} - SPF_t$.

- Log-likelihood: **−222.664**
- AIC: **447.329**, BIC: **450.700**
- Residual skewness: **−0.713**, excess kurtosis: **7.301**

ARX(1,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0824} \; \underset{(0.112)}{+0.3005\,\pi_t} \; \underset{(0.167)}{+0.7337\,SPF_t} \tag{2}$$

- Log-likelihood: **−207.381**
- AIC: **420.762**, BIC: **430.874**
- Residual skewness: **−1.047**, excess kurtosis: **8.210**

ARX(2,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0897} \; \underset{(0.098)}{+0.2892\,\pi_t} \; \underset{(0.126)}{+0.0834\,\pi_{t-1}} \; \underset{(0.251)}{+0.6385\,SPF_t} \tag{3}$$

- Log-likelihood: **−206.810**
- AIC: **421.619**, BIC: **435.102**
- Residual skewness: **−1.191**, excess kurtosis: **9.050**

ARX(2,2):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0856} \; \underset{(0.106)}{+0.2992\,\pi_t} \; \underset{(0.125)}{+0.0914\,\pi_{t-1}} \; \underset{(0.433)}{+0.4084\,SPF_t} \; \underset{(0.297)}{+0.2136\,SPF_{t-1}} \tag{4}$$

- Log-likelihood: **−206.660**
- AIC: **423.319**, BIC: **440.172**
- Residual skewness: **−1.212**, excess kurtosis: **9.129**

2.b. Monthly Results

The same four mean models estimated on the monthly effective sample (1969M2–2026M5, 688 observations; see the Monthly Data section of the Data Summary) are reported in

`OLS_Results_Monthly.md`, generated by `LinearRegressions_Monthly.py` with HAC(12) standard errors. At the monthly frequency ARX(1,1) is preferred by both AIC and BIC; the additional inflation and SPF lags in ARX(2,1) and ARX(2,2) are individually insignificant.

3. OLS Results (Monthly)

Estimation results of four mean models on the monthly effective sample (1969-02–2026-05, 688 observations) with HAC(12) standard errors in parentheses below each estimate. (The log-likelihood is computed under the Gaussian assumption.)

Constant:

$$\hat{\pi}_{t+1} = SPF_t \tag{5}$$

No estimated coefficients. Residuals $\mu_{t+1} = \pi_{t+1} - SPF_t$.

- Log-likelihood: **–96.715**
- AIC: **195.430**, BIC: **199.963**
- Residual skewness: –**0.413**, excess kurtosis: **6.728**

ARX(1,1):

$$\hat{\pi}_{t+1} = \underset{(0.020)}{0.0195} \; \underset{(0.054)}{+0.4736\,\pi_t} \; \underset{(0.099)}{+0.5577\,SPF_t} \tag{6}$$

- Log-likelihood: **1.564**
- AIC: **2.871**, BIC: **16.473**
- Residual skewness: –**0.074**, excess kurtosis: **5.909**

ARX(2,1):

$$\hat{\pi}_{t+1} = \underset{(0.020)}{0.0195} \; \underset{(0.075)}{+0.4843\,\pi_t} \; \underset{(0.077)}{-0.0242\,\pi_{t-1}} \; \underset{(0.092)}{+0.5740\,SPF_t} \tag{7}$$

- Log-likelihood: **1.757**
- AIC: **4.486**, BIC: **22.622**
- Residual skewness: –**0.060**, excess kurtosis: **5.950**

ARX(2,2):

$$\hat{\pi}_{t+1} = \underset{(0.021)}{0.0178} \; \underset{(0.075)}{+0.4874\,\pi_t} \; \underset{(0.079)}{-0.0198\,\pi_{t-1}} \; \underset{(0.417)}{+0.0108\,SPF_t} \; \underset{(0.392)}{+0.5601\,SPF_{t-1}} \tag{8}$$

- Log-likelihood: **2.663**
- AIC: **4.674**, BIC: **27.343**
- Residual skewness: –**0.054**, excess kurtosis: **5.920**

4. GARCH Family

This section presents three GARCH-type volatility models — GARCH, GJR-GARCH, and EGARCH — each combined with one of three conditional distributions for residuals: normal, Student's t, and a two-component Gaussian mixture.

Let $\{u_t\}_{t=1}^{T}$ denote the residual process, and let $\mathcal{F}_{t-1}$ denote the information set available at time $t-1$. I assume

$$u_t = \sigma_t \, z_t, \qquad z_t \mid \mathcal{F}_{t-1} \overset{\text{i. i. d.}}{\sim} D(0,1), \tag{9}$$

where $\sigma_t^2 = \operatorname{Var}(u_t \mid \mathcal{F}_{t-1})$ is the conditional variance and $D(0,1)$ is a standardized distribution (normal, Student's t, or Gaussian mixture) with zero mean and unit variance. Throughout, $z_t = u_t / \sigma_t$ denotes the standardized residual and θ denotes the full parameter vector (mean process, volatility process parameters together with any distributional parameters).

4.a. Volatility Recursion Specifications

GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \, u_{t-i}^2 + \sum_{k=1}^{q} \beta_k \, \sigma_{t-k}^2, \tag{10}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\beta_k \geq 0$, and $\sum_i \alpha_i + \sum_k \beta_k < 1$ for stationarity.

GJR-GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \left(u_{t-i}^{+}\right)^2 + \sum_{j=1}^{p} \gamma_j \left(u_{t-j}^{-}\right)^2 + \sum_{k=1}^{q} \beta_k \, \sigma_{t-k}^2, \tag{11}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\gamma_j \geq 0$, and $\beta_k \geq 0$. The asymmetry coefficient γ_j captures the leverage effect: when $\alpha_j > \gamma_j$, positive shocks raise future volatility more than negative shocks of equal magnitude.

Under the assumption that z_t is symmetric

$$\mathbb{E}\left[\left(u_{t-i}^{+}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \mathbb{E}\left[\left(u_{t-i}^{-}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \frac{1}{2} \sigma_{t-i}^2 \tag{12}$$

, the stationarity condition is

$$\frac{1}{2} \sum_{i=1}^{p} \alpha_i + \frac{1}{2} \sum_{j=1}^{p} \gamma_j + \sum_{k=1}^{q} \beta_k < 1. \tag{13}$$

EGARCH(p, q):

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \left(\mid z_{t-i} \mid - \sqrt{\frac{2}{\pi}} \right) + \sum_{j=1}^{p} \gamma_j \, z_{t-j} + \sum_{k=1}^{q} \beta_k \ln \sigma_{t-k}^2, \tag{14}$$

where $z_t = u_t / \sigma_t$ is the standardized residual. The term α_i captures the magnitude effect (response to the size of shocks) while γ_j captures the sign / leverage effect. Modeling $\ln \sigma_t^2$ guarantees positivity of σ_t^2 without sign restrictions on $\left(\alpha_i, \gamma_j, \beta_k\right)$, and the recursion is covariance-stationary in $\ln \sigma_t^2$ whenever

$$\sum_{k=1}^{q} \beta_k < 1. \tag{15}$$

4.b. Estimation Initialization And Collection

The GARCH-family estimation is executed by the `arch` package. For each distribution, mean process, volatility family, and order, the script tries the package default optimizer start and additional randomized feasible starts. For ARX mean specifications, the mean-parameter block in randomized starts is drawn from OLS-centered intervals, $\hat{\theta}_j \pm 2\, SE\left(\hat{\theta}_j\right)$, while volatility and distribution parameters are randomized over feasible regions. The current reporting run uses 50 starts per model combination. The preferred reported estimate is the highest log-likelihood estimate among starts that both converged according to the optimizer and passed the relevant stationarity proxy with a small numerical margin below one (`1e-6`). If none of the 50 starts passes all checks, the script reports the best optimizer-converged attempted log likelihood for that model combination and flags the row in the CSV `selection_status` column. Attempt-level restart diagnostics are also saved for audit. Conditional-variance recursion starts use the default initialization implemented by the `arch` package.

4.c. Log-Likelihood Specifications

For each candidate distribution we give (i) the standardized density of z_t, (ii) the conditional density of $u_t \mid \mathcal{F}_{t-1}$ obtained via the change of variables $u_t = \sigma_t z_t$ (which contributes a Jacobian factor $1/\sigma_t$), (iii) the per-observation log-likelihood $\ell_t(\theta)$, and (iv) the sample log-likelihood

$$\ln L(\theta) = \sum_{t=1}^{T} \ell_t(\theta). \tag{16}$$

4.c.i. *Normal distribution:*

The standardized innovation satisfies $z_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, 1)$ with density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\}. \tag{17}$$

By the change of variables $u_t = \sigma_t z_t$, the conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} \phi\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sqrt{2\pi\, \sigma_t^2}} \exp\left\{-\frac{u_t^2}{2\, \sigma_t^2}\right\}. \tag{18}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\frac{1}{2}\left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right), \tag{19}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\frac{1}{2} \sum_{t=1}^{T} \left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right). \tag{20}$$

4.c.ii. *Student's t distribution:*

Let $\nu > 2$ denote the degrees-of-freedom parameter. The standardized Student's t density (rescaled to unit variance) is

$$f_\nu(z) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)}} \left(1 + \frac{z^2}{\nu-2}\right)^{-\frac{\nu+1}{2}}, \tag{21}$$

where $\Gamma(\cdot)$ is the gamma function. The restriction $\nu > 2$ ensures finite variance. The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} f_\nu\left(\frac{u_t}{\sigma_t}\right) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)\,\sigma_t^2}} \left(1 + \frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right)^{-\frac{\nu+1}{2}}. \tag{22}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = \ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\sigma_t^2\ \right) - \frac{\nu+1}{2}\ln\left(1 + \frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right), \tag{23}$$

and the sample log-likelihood is

$$\ln L(\theta) = T\left[\ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\ \right)\right] - \frac{1}{2}\sum_{t=1}^{T}\ln\sigma_t^2 - \frac{\nu+1}{2}\sum_{t=1}^{T}\ln\left(1 + \frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right). \tag{24}$$

4.c.iii. *Gaussian mixture distribution:*

The standardized innovation follows a two-component Gaussian mixture. The free parameters are (p_1, μ_1, σ_1^2):

$$f_{\text{mix}}(z) = p_1\,\phi(z; \mu_1, \sigma_1^2) + p_2\,\phi(z; \mu_2, \sigma_2^2), \qquad \phi(z; \mu, s^2) = \frac{1}{\sqrt{2\pi s^2}}\exp\left\{-\frac{(z-\mu)^2}{2s^2}\right\}. \tag{25}$$

To enforce $\mathbb{E}[z_t] = 0$ and $\mathrm{Var}(z_t) = 1$, the remaining parameters are determined by

$$p_2 = 1 - p_1, \qquad \mu_2 = -\frac{p_1\mu_1}{p_2}, \qquad \sigma_2^2 = \frac{1 - p_1(\sigma_1^2 + \mu_1^2) - p_2\mu_2^2}{p_2}. \tag{26}$$

This leaves three free parameters: (p_1, μ_1, σ_1^2).

Hierarchical representation. The mixture admits the latent-variable form $S \sim \text{Bernoulli}(p_1)$, $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, with $z = X_1$ if $S = 1$ and $z = X_2$ otherwise. As a consequence, moments and the CDF decompose as

$$\mathbb{E}[z^n] = p_1\,\mathbb{E}[X_1^n] + p_2\,\mathbb{E}[X_2^n], \qquad F_{\text{mix}}(a) = p_1\,\Phi\left(\frac{a-\mu_1}{\sigma_1}\right) + p_2\,\Phi\left(\frac{a-\mu_2}{\sigma_2}\right), \tag{27}$$

where Φ is the standard normal CDF.

The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} f_{\text{mix}}\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sigma_t}\left[p_1\,\phi(z_t; \mu_1, \sigma_1^2) + p_2\,\phi(z_t; \mu_2, \sigma_2^2)\right], \qquad z_t = \frac{u_t}{\sigma_t}. \tag{28}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\ln\sigma_t + \ln\left[p_1\,\phi(z_t; \mu_1, \sigma_1^2) + p_2\,\phi(z_t; \mu_2, \sigma_2^2)\right], \tag{29}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\sum_{t=1}^{T} \ln \sigma_t + \sum_{t=1}^{T} \ln\big[\, p_1\, \phi(z_t; \mu_1, \sigma_1^2) + p_2\, \phi(z_t; \mu_2, \sigma_2^2) \big]. \tag{30}$$

4.d. Monthly Estimation

The same estimation menu (three volatility families, four mean processes, three residual distributions, 50 starts per combination) can be run on the monthly effective sample (1969M2–2026M5, 688 observations):

```
python estimate_garch_results.py \
    --data-path ../DataSummary/Aggregate_CPI_inflation_Monthly.pkl \
    --output-dir results_garch_distributions_monthly
```

Monthly outputs are stored in `results_garch_distributions_monthly/`, parallel to the quarterly `results_garch_distributions/`. All model settings are identical across the two frequencies.

5. Model Selection Report

For each model combination, the reported row uses the highest log-likelihood estimate that passes optimizer convergence and stationarity checks across 50 starts. If no start passes all checks, the reported row uses the best optimizer-converged attempted log likelihood and is flagged in the CSV `selection_status` column.

Table 2: Model selection by AIC: GARCH vs GJR vs EGARCH[2]

Distribution	GARCH Mean	GARCH Vol	GARCH AIC	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	ARX(2,1)	(2,1)	387.2314	ARX(1,1)	(2,1)	372.5785	ARX(1,1)	(1,2)	358.6843
Student's t	ARX(2,1)	(1,1)	361.3528	ARX(2,1)	(2,1)	356.9953	ARX(2,1)	(1,2)	351.8931
Gaussian mixture	ARX(1,1)	(2,1)	367.1233	ARX(2,1)	(2,1)	358.4230	ARX(2,1)	(1,1)	352.7338

Table 3: Model selection by BIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH BIC	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	Constant	(2,1)	401.2826	Constant	(2,1)	396.7277	ARX(1,1)	(1,2)	385.6494
Student's t	Constant	(2,1)	386.6231	Constant	(1,1)	384.2364	Constant	(1,1)	383.5813
Gaussian mixture	ARX(1,1)	(2,1)	400.8297	ARX(1,1)	(1,1)	394.1515	Constant	(1,1)	385.2612

5.a. *Best Models by Criterion and Volatility Family*

For each criterion and each volatility family, the selected model is the best across mean-process choices, orders, and distributions using stable & successful fits. The main result tables contain 144 reported model combinations.

5.a.i. *AIC:*

GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M073`	Student's t	ARX(2,1)	GARCH(1,1)	−172.6764	361.3528	388.3179

Mean process:

$$\hat{\pi}_{t+1} = 0.0876 + 0.2637\,\pi_t + 0.1716\,\pi_{t-1} + 0.5096\,SPF_t + \mu_{t+1} \tag{31}$$

Volatility process:

[2]These numbers differ from earlier GARCH tables because earlier code used inconsistent effective sample sizes across model settings due to a sample-trimming error. The correction is discussed in the Data Summary effective-sample section. The current estimates use the common **1969Q2–2022Q4** sample with **215 observations**.

$$\hat{\sigma}_t^2 = 0.0725 + 0.5202\, u_{t-1}^2 + 0.4294\, \sigma_{t-1}^2 \tag{32}$$

Distribution parameters:

Parameter	Estimate
ν	4.3024

Parameter	Coef	Std Err	t-value	p-value
c	0.0876	0.0777	1.1279	0.2593
ρ_1	0.2637	0.0912	2.8914	0.0038
ρ_2	0.1716	0.0975	1.7600	0.0784
ϕ_1	0.5096	0.1487	3.4274	0.0006
ω	0.0725	0.0385	1.8834	0.0596
α_1	0.5202	0.2424	2.1465	0.0318
β_1	0.4294	0.1425	3.0126	0.0026
ν	4.3024	1.4647	2.9373	0.0033

GJR-GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M080`	Student's t	ARX(2,1)	GJR-GARCH(2,1)	−167.4976	356.9953	394.0723

Mean process:

$$\hat{\pi}_{t+1} = 0.0901 + 0.2045\, \pi_t + 0.1144\, \pi_{t-1} + 0.6522\, SPF_t + \mu_{t+1} \tag{33}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.1096 + 0.4005\, (u_{t-1}^+)^2 + 0.2848\, (u_{t-1}^-)^2 + 0.6415\, (u_{t-2}^+)^2 + 0.0000\, (u_{t-2}^-)^2 + 0.1559\, \sigma_{t-1}^2 \tag{34}$$

Distribution parameters:

Parameter	Estimate
ν	5.3421

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.0901	0.0838	1.0747	0.2825
ρ_1	0.2045	0.0960	2.1292	0.0332
ρ_2	0.1144	0.1043	1.0969	0.2727
ϕ_1	0.6522	0.1740	3.7481	0.0002
ω	0.1096	0.0397	2.7593	0.0058

Parameter	Coef	Std Err	t-value	p-value
α_1	0.4005	0.2194	1.8250	0.0680
$\gamma_1 - \alpha_1$	−0.1157	0.2773	−0.4173	0.6765
α_2	0.6415	0.3571	1.7962	0.0725
$\gamma_2 - \alpha_2$	−0.6415	0.3194	−2.0082	0.0446
β_1	0.1559	0.1500	1.0399	0.2984
ν	5.3421	1.9529	2.7354	0.0062

EGARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M078`	Student's t	ARX(2,1)	EGARCH(1,2)	−165.9465	351.8931	385.5994

Mean process:

$$\hat{\pi}_{t+1} = 0.0074 + 0.1430\,\pi_t + 0.1015\,\pi_{t-1} + 0.8896\,SPF_t + \mu_{t+1} \tag{35}$$

Volatility process:

$$\ln \hat{\sigma}_t^2 = -0.0740 - 0.3421\left(\mid z_{t-1} \mid - \sqrt{2/\pi}\right) + 0.2355\,z_{t-1} + 0.0412 \ln \sigma_{t-1}^2 + 0.9200 \ln \sigma_{t-2}^2 \tag{36}$$

Distribution parameters:

Parameter	Estimate
ν	4.5989

Parameter	Coef	Std Err	t-value	p-value
c	0.0074	0.0002	33.8794	0.0000
ρ_1	0.1430	0.0016	91.7625	0.0000
ρ_2	0.1015	0.0016	63.3776	0.0000
ϕ_1	0.8896	0.0000	2430766.0218	0.0000
ω	−0.0740	0.0000	−50401.4684	0.0000
α_1	−0.3421	0.0001	−2676.8779	0.0000
γ_1	0.2355	0.0010	241.7362	0.0000
β_1	0.0412	0.0002	186.9354	0.0000
β_2	0.9200	0.0000	728877.9846	0.0000
ν	4.5989	0.2382	19.3104	0.0000

5.a.ii. *BIC:*

GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M055`	Student's t	Constant	GARCH(2,1)	−179.8850	369.7699	386.6231

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{37}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\hat{\sigma}_t^2 = 0.1192 + 0.4570\, u_{t-1}^2 + 0.5209\, u_{t-2}^2 + 0.0000\, \sigma_{t-1}^2 \tag{38}$$

Distribution parameters:

Parameter	Estimate
ν	4.9416

Parameter	Coef	Std Err	t-value	p-value
ω	0.1192	0.0417	2.8561	0.0043
α_1	0.4570	0.1774	2.5755	0.0100
α_2	0.5209	0.2133	2.4419	0.0146
β_1	0.0000	0.1189	0.0000	1.0000
ν	4.9416	1.7105	2.8890	0.0039

GJR-GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M050`	Student's t	Constant	GJR-GARCH(1,1)	−178.6916	367.3832	384.2364

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{39}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\hat{\sigma}_t^2 = 0.0636 + 0.7969\, (u_{t-1}^+)^2 + 0.1469\, (u_{t-1}^-)^2 + 0.4353\, \sigma_{t-1}^2 \tag{40}$$

Distribution parameters:

Parameter	Estimate
ν	4.7915

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
Parameter	Coef	Std Err	t-value	p-value
ω	0.0636	0.0255	2.4924	0.0127
α_1	0.7969	0.2566	3.1059	0.0019
$\gamma_1 - \alpha_1$	−0.6499	0.2320	−2.8010	0.0051
β_1	0.4353	0.1168	3.7261	0.0002
ν	4.7915	1.5515	3.0883	0.0020

EGARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M051`	Student's t	Constant	EGARCH(1,1)	−178.3641	366.7281	383.5813

Mean process:

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{41}$$

No mean-process coefficients are estimated in this anchored specification.

Volatility process:

$$\ln \hat{\sigma}_t^2 = -0.2496 + 0.5580 \left(\mid z_{t-1} \mid - \sqrt{2/\pi} \right) + 0.2647\, z_{t-1} + 0.7897 \ln \sigma_{t-1}^2 \tag{42}$$

Distribution parameters:

Parameter	Estimate
ν	5.0826

Parameter	Coef	Std Err	t-value	p-value
ω	−0.2496	0.1063	−2.3491	0.0188
α_1	0.5580	0.1395	4.0011	0.0001
γ_1	0.2647	0.0782	3.3870	0.0007
β_1	0.7897	0.0610	12.9555	0.0000
ν	5.0826	1.7271	2.9428	0.0033

5.b. All Model Results

Distribution	Mean Process	Volatility Process	LogLik	AIC	BIC
Normal	Constant	GARCH(1,1)	−198.2015	402.4029	412.5148
Normal	Constant	GARCH(1,2)	−198.2015	404.4029	417.8855
Normal	Constant	GARCH(2,1)	−189.9000	387.8000	401.2826
Normal	Constant	GARCH(2,2)	−189.9000	389.8000	406.6532
Normal	Constant	GJR-GARCH(1,1)	−189.8927	387.7854	401.2680
Normal	Constant	GJR-GARCH(1,2)	−189.8927	389.7854	406.6386

Normal	Constant	GJR-GARCH(2,1)	−182.2520	376.5039	396.7277
Distribution	Mean Process	Volatility Process	LogLik	AIC	BIC
Normal	Constant	GJR-GARCH(2,2)	−182.2520	378.5039	402.0984
Normal	Constant	EGARCH(1,1)	−188.9696	385.9392	399.4217
Normal	Constant	EGARCH(1,2)	−188.9696	387.9392	404.7924
Normal	Constant	EGARCH(2,1)	−182.9516	377.9032	398.1271
Normal	Constant	EGARCH(2,2)	−182.9516	379.9032	403.4977
Normal	ARX(1,1)	GARCH(1,1)	−191.0320	394.0640	414.2879
Normal	ARX(1,1)	GARCH(1,2)	−191.0320	396.0640	419.6585
Normal	ARX(1,1)	GARCH(2,1)	−186.6387	387.2773	410.8718
Normal	ARX(1,1)	GARCH(2,2)	−186.6387	389.2773	416.2424
Normal	ARX(1,1)	GJR-GARCH(1,1)	−183.8229	381.6458	405.2403
Normal	ARX(1,1)	GJR-GARCH(1,2)	−183.8229	383.6458	410.6109
Normal	ARX(1,1)	GJR-GARCH(2,1)	−177.2892	372.5785	402.9142
Normal	ARX(1,1)	GJR-GARCH(2,2)	−177.2892	374.5785	408.2849
Normal	ARX(1,1)	EGARCH(1,1)	−183.7595	381.5190	405.1134
Normal	ARX(1,1)	EGARCH(1,2)	−171.3421	358.6843	385.6494
Normal	ARX(1,1)	EGARCH(2,1)	−178.1375	374.2750	404.6107
Normal	ARX(1,1)	EGARCH(2,2)	−171.2540	362.5081	396.2144
Normal	ARX(2,1)	GARCH(1,1)	−188.8527	391.7055	415.2999
Normal	ARX(2,1)	GARCH(1,2)	−188.8527	393.7055	420.6706
Normal	ARX(2,1)	GARCH(2,1)	−185.6157	387.2314	414.1965
Normal	ARX(2,1)	GARCH(2,2)	−185.6157	389.2314	419.5672
Normal	ARX(2,1)	GJR-GARCH(1,1)	−183.7897	383.5794	410.5445
Normal	ARX(2,1)	GJR-GARCH(1,2)	−183.7897	385.5794	415.9151
Normal	ARX(2,1)	GJR-GARCH(2,1)	−176.6378	373.2755	406.9819
Normal	ARX(2,1)	GJR-GARCH(2,2)	−176.6378	375.2755	412.3525
Normal	ARX(2,1)	EGARCH(1,1)	−183.4410	382.8820	409.8471
Normal	ARX(2,1)	EGARCH(1,2)	−171.1430	360.2860	390.6217
Normal	ARX(2,1)	EGARCH(2,1)	−177.9523	375.9046	409.6110
Normal	ARX(2,1)	EGARCH(2,2)	−177.9523	377.9046	414.9816
Normal	ARX(2,2)	GARCH(1,1)	−188.8355	393.6711	420.6362
Normal	ARX(2,2)	GARCH(1,2)	−188.8355	395.6711	426.0068
Normal	ARX(2,2)	GARCH(2,1)	−185.6074	389.2147	419.5505
Normal	ARX(2,2)	GARCH(2,2)	−185.6074	391.2147	424.9211
Normal	ARX(2,2)	GJR-GARCH(1,1)	−183.7821	385.5642	415.9000
Normal	ARX(2,2)	GJR-GARCH(1,2)	−183.7821	387.5642	421.2706

Normal	ARX(2,2)	GJR-GARCH(2,1)	−176.5660	375.1321	412.2091
Distribution	Mean Process	Volatility Process	LogLik	AIC	BIC
Normal	ARX(2,2)	GJR-GARCH(2,2)	−176.5660	377.1321	417.5797
Normal	ARX(2,2)	EGARCH(1,1)	−182.8681	383.7363	414.0720
Normal	ARX(2,2)	EGARCH(1,2)	−182.8681	385.7363	419.4427
Normal	ARX(2,2)	EGARCH(2,1)	−177.5298	377.0595	414.1365
Normal	ARX(2,2)	EGARCH(2,2)	−177.5298	379.0595	419.5072
Student's t	Constant	GARCH(1,1)	−181.9942	371.9883	385.4709
Student's t	Constant	GARCH(1,2)	−181.9942	373.9883	390.8415
Student's t	Constant	GARCH(2,1)	−179.8850	369.7699	386.6231
Student's t	Constant	GARCH(2,2)	−179.8850	371.7699	391.9938
Student's t	Constant	GJR-GARCH(1,1)	−178.6916	367.3832	384.2364
Student's t	Constant	GJR-GARCH(1,2)	−178.6916	369.3832	389.6070
Student's t	Constant	GJR-GARCH(2,1)	−174.8160	363.6320	387.2265
Student's t	Constant	GJR-GARCH(2,2)	−174.8160	365.6320	392.5972
Student's t	Constant	EGARCH(1,1)	−178.3641	366.7281	383.5813
Student's t	Constant	EGARCH(1,2)	−178.3641	368.7281	388.9520
Student's t	Constant	EGARCH(2,1)	−175.5435	365.0870	388.6815
Student's t	Constant	EGARCH(2,2)	−175.5435	367.0870	394.0521
Student's t	ARX(1,1)	GARCH(1,1)	−174.9174	363.8348	387.4293
Student's t	ARX(1,1)	GARCH(1,2)	−174.9174	365.8348	392.7999
Student's t	ARX(1,1)	GARCH(2,1)	−173.7890	363.5780	390.5431
Student's t	ARX(1,1)	GARCH(2,2)	−173.7890	365.5780	395.9138
Student's t	ARX(1,1)	GJR-GARCH(1,1)	−171.2540	358.5080	385.4731
Student's t	ARX(1,1)	GJR-GARCH(1,2)	−171.2540	360.5080	390.8437
Student's t	ARX(1,1)	GJR-GARCH(2,1)	−168.5408	357.0815	390.7879
Student's t	ARX(1,1)	GJR-GARCH(2,2)	−168.5408	359.0815	396.1586
Student's t	ARX(1,1)	EGARCH(1,1)	−171.0753	358.1506	385.1157
Student's t	ARX(1,1)	EGARCH(1,2)	−171.0753	360.1506	390.4864
Student's t	ARX(1,1)	EGARCH(2,1)	−168.3693	356.7385	390.4449
Student's t	ARX(1,1)	EGARCH(2,2)	−168.3693	358.7385	395.8155
Student's t	ARX(2,1)	GARCH(1,1)	−172.6764	361.3528	388.3179
Student's t	ARX(2,1)	GARCH(1,2)	−172.6764	363.3528	393.6885
Student's t	ARX(2,1)	GARCH(2,1)	−171.9737	361.9474	392.2832
Student's t	ARX(2,1)	GARCH(2,2)	−171.6977	363.3953	397.1017
Student's t	ARX(2,1)	GJR-GARCH(1,1)	−170.0718	358.1436	388.4794
Student's t	ARX(2,1)	GJR-GARCH(1,2)	−170.0718	360.1436	393.8500

Distribution	Mean Process	Volatility Process	LogLik	AIC	BIC
Student's t	ARX(2,1)	GJR-GARCH(2,1)	−167.4976	356.9953	394.0723
Student's t	ARX(2,1)	GJR-GARCH(2,2)	−167.4976	358.9953	399.4429
Student's t	ARX(2,1)	EGARCH(1,1)	−169.7106	357.4213	387.7570
Student's t	ARX(2,1)	EGARCH(1,2)	−165.9465	351.8931	385.5994
Student's t	ARX(2,1)	EGARCH(2,1)	−167.5810	357.1621	394.2391
Student's t	ARX(2,1)	EGARCH(2,2)	−167.5810	359.1621	399.6098
Student's t	ARX(2,2)	GARCH(1,1)	−172.5010	363.0019	393.3377
Student's t	ARX(2,2)	GARCH(1,2)	−172.5010	365.0019	398.7083
Student's t	ARX(2,2)	GARCH(2,1)	−171.8786	363.7573	397.4636
Student's t	ARX(2,2)	GARCH(2,2)	−171.6518	365.3035	402.3805
Student's t	ARX(2,2)	GJR-GARCH(1,1)	−169.9900	359.9800	393.6864
Student's t	ARX(2,2)	GJR-GARCH(1,2)	−169.9900	361.9800	399.0571
Student's t	ARX(2,2)	GJR-GARCH(2,1)	−167.3944	358.7887	399.2364
Student's t	ARX(2,2)	GJR-GARCH(2,2)	−167.3944	360.7887	404.6070
Student's t	ARX(2,2)	EGARCH(1,1)	−169.4442	358.8885	392.5948
Student's t	ARX(2,2)	EGARCH(1,2)	−169.4442	360.8885	397.9655
Student's t	ARX(2,2)	EGARCH(2,1)	−167.2956	358.5911	399.0388
Student's t	ARX(2,2)	EGARCH(2,2)	−167.2956	360.5911	404.4094
Gaussian mixture	Constant	GARCH(1,1)	−179.4873	370.9746	391.1984
Gaussian mixture	Constant	GARCH(1,2)	−179.4873	372.9746	396.5690
Gaussian mixture	Constant	GARCH(2,1)	−177.7161	369.4323	393.0268
Gaussian mixture	Constant	GARCH(2,2)	−177.7161	371.4323	398.3974
Gaussian mixture	Constant	GJR-GARCH(1,1)	−175.2123	364.4246	388.0190
Gaussian mixture	Constant	GJR-GARCH(1,2)	−175.2123	366.4246	393.3897
Gaussian mixture	Constant	GJR-GARCH(2,1)	−171.8210	361.6420	391.9777
Gaussian mixture	Constant	GJR-GARCH(2,2)	−171.8210	363.6420	397.3484
Gaussian mixture	Constant	EGARCH(1,1)	−173.8334	361.6667	385.2612
Gaussian mixture	Constant	EGARCH(1,2)	−173.8334	363.6667	390.6318
Gaussian mixture	Constant	EGARCH(2,1)	−170.7788	359.5577	389.8934
Gaussian mixture	Constant	EGARCH(2,2)	−170.7788	361.5577	395.2641
Gaussian mixture	ARX(1,1)	GARCH(1,1)	−174.5852	367.1704	397.5061
Gaussian mixture	ARX(1,1)	GARCH(1,2)	−174.5852	369.1704	402.8768
Gaussian mixture	ARX(1,1)	GARCH(2,1)	−173.5617	367.1233	400.8297
Gaussian mixture	ARX(1,1)	GARCH(2,2)	−173.5617	369.1233	406.2004
Gaussian mixture	ARX(1,1)	GJR-GARCH(1,1)	−170.2226	360.4452	394.1515
Gaussian mixture	ARX(1,1)	GJR-GARCH(1,2)	−170.2226	362.4452	399.5222

Distribution	Mean Process	Volatility Process	LogLik	AIC	BIC
Gaussian mixture	ARX(1,1)	GJR-GARCH(2,1)	−167.4469	358.8939	399.3415
Gaussian mixture	ARX(1,1)	GJR-GARCH(2,2)	−167.4469	360.8939	404.7122
Gaussian mixture	ARX(1,1)	EGARCH(1,1)	−169.4422	358.8844	392.5908
Gaussian mixture	ARX(1,1)	EGARCH(1,2)	−169.4422	360.8844	397.9614
Gaussian mixture	ARX(1,1)	EGARCH(2,1)	−166.2197	356.4394	396.8871
Gaussian mixture	ARX(1,1)	EGARCH(2,2)	−166.2197	358.4394	402.2577
Gaussian mixture	ARX(2,1)	GARCH(1,1)	−171.3839	362.7678	396.4742
Gaussian mixture	ARX(2,1)	GARCH(1,2)	−171.3839	364.7678	401.8449
Gaussian mixture	ARX(2,1)	GARCH(2,1)	−170.7809	363.5618	400.6389
Gaussian mixture	ARX(2,1)	GARCH(2,2)	−170.6186	365.2371	405.6848
Gaussian mixture	ARX(2,1)	GJR-GARCH(1,1)	−168.6131	359.2261	396.3031
Gaussian mixture	ARX(2,1)	GJR-GARCH(1,2)	−168.6131	361.2261	401.6738
Gaussian mixture	ARX(2,1)	GJR-GARCH(2,1)	−166.2115	358.4230	402.2413
Gaussian mixture	ARX(2,1)	GJR-GARCH(2,2)	−166.2115	360.4230	407.6120
Gaussian mixture	ARX(2,1)	EGARCH(1,1)	−165.3669	352.7338	389.8108
Gaussian mixture	ARX(2,1)	EGARCH(1,2)	−164.6031	353.2062	393.6539
Gaussian mixture	ARX(2,1)	EGARCH(2,1)	−165.2689	356.5377	400.3560
Gaussian mixture	ARX(2,1)	EGARCH(2,2)	−163.6882	355.3765	402.5654
Gaussian mixture	ARX(2,2)	GARCH(1,1)	−171.1994	364.3989	401.4759
Gaussian mixture	ARX(2,2)	GARCH(1,2)	−171.1994	366.3989	406.8465
Gaussian mixture	ARX(2,2)	GARCH(2,1)	−170.7011	365.4022	405.8499
Gaussian mixture	ARX(2,2)	GARCH(2,2)	−174.8814	375.7628	419.5811
Gaussian mixture	ARX(2,2)	GJR-GARCH(1,1)	−168.4801	360.9602	401.4079
Gaussian mixture	ARX(2,2)	GJR-GARCH(1,2)	−168.4801	362.9602	406.7785
Gaussian mixture	ARX(2,2)	GJR-GARCH(2,1)	−166.0585	360.1169	407.3059
Gaussian mixture	ARX(2,2)	GJR-GARCH(2,2)	−166.0585	362.1169	412.6765
Gaussian mixture	ARX(2,2)	EGARCH(1,1)	−167.3628	358.7257	399.1733
Gaussian mixture	ARX(2,2)	EGARCH(1,2)	−166.4468	358.8937	402.7120
Gaussian mixture	ARX(2,2)	EGARCH(2,1)	−162.5823	353.1646	400.3535
Gaussian mixture	ARX(2,2)	EGARCH(2,2)	−164.5054	359.0108	409.5704

6. Model Selection Report

For each model combination, the reported row uses the highest log-likelihood estimate that passes optimizer convergence and stationarity checks across 50 starts. If no start passes all checks, the reported row uses the best optimizer-converged attempted log likelihood and is flagged in the CSV `selection_status` column.

Table 2: Model selection by AIC: GARCH vs GJR vs EGARCH[3]

Distribution	GARCH Mean	GARCH Vol	GARCH AIC	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	ARX(1,1)	(1,1)	−140.5477	ARX(1,1)	(2,2)	−143.8422	ARX(1,1)	(2,1)	−135.8873
Student's t	ARX(1,1)	(1,1)	−186.3088	ARX(1,1)	(1,1)	−188.2854	ARX(1,1)	(1,1)	−180.8314
Gaussian mixture	ARX(1,1)	(1,1)	−185.3003	ARX(1,1)	(1,1)	−187.6112	ARX(1,1)	(1,1)	−181.8980

Table 3: Model selection by BIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH BIC	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	ARX(1,1)	(1,1)	−113.3450	ARX(1,1)	(1,1)	−108.7612	ARX(1,1)	(1,1)	−95.4457
Student's t	ARX(1,1)	(1,1)	−154.5723	ARX(1,1)	(1,1)	−152.0151	ARX(1,1)	(1,1)	−144.5611
Gaussian mixture	ARX(1,1)	(1,1)	−144.4962	ARX(1,1)	(1,1)	−142.2734	ARX(1,1)	(1,1)	−136.5601

6.a. All Model Results

Distribution	Mean Process	Volatility Process	Best LogLik	AIC	BIC
Normal	Constant	GARCH(1,1)	21.1614	−36.3228	−22.7215
Normal	Constant	GARCH(1,2)	22.6158	−37.2315	−19.0963
Normal	Constant	GARCH(2,1)	21.1614	−34.3228	−16.1877
Normal	Constant	GARCH(2,2)	22.6158	−35.2315	−12.5626
Normal	Constant	GJR-GARCH(1,1)	21.3677	−34.7353	−16.6002
Normal	Constant	GJR-GARCH(1,2)	22.9861	−35.9722	−13.3033
Normal	Constant	GJR-GARCH(2,1)	21.6293	−31.2587	−4.0559
Normal	Constant	GJR-GARCH(2,2)	25.9373	−37.8747	−6.1381
Normal	Constant	EGARCH(1,1)	16.5685	−25.1371	−7.0019
Normal	Constant	EGARCH(1,2)	19.0223	−28.0446	−5.3756

[3]These numbers differ from earlier GARCH tables because earlier code used inconsistent effective sample sizes across model settings due to a sample-trimming error. The correction is discussed in the Data Summary effective-sample section. The current estimates use a common effective sample with **688 observations**.

Distribution	Mean Process	Volatility Process	Best LogLik	AIC	BIC
Normal	Constant	EGARCH(2,1)	25.9907	−39.9814	−12.7787
Normal	Constant	EGARCH(2,2)	25.9907	−37.9814	−6.2449
Normal	ARX(1,1)	GARCH(1,1)	76.2739	−140.5477	−113.3450
Normal	ARX(1,1)	GARCH(1,2)	76.2746	−138.5492	−106.8127
Normal	ARX(1,1)	GARCH(2,1)	76.2739	−138.5477	−106.8112
Normal	ARX(1,1)	GARCH(2,2)	76.3475	−136.6950	−100.4247
Normal	ARX(1,1)	GJR-GARCH(1,1)	77.2489	−140.4978	−108.7612
Normal	ARX(1,1)	GJR-GARCH(1,2)	77.2499	−138.4998	−102.2295
Normal	ARX(1,1)	GJR-GARCH(2,1)	79.6561	−141.3123	−100.5082
Normal	ARX(1,1)	GJR-GARCH(2,2)	81.9211	−143.8422	−98.5043
Normal	ARX(1,1)	EGARCH(1,1)	70.5911	−127.1822	−95.4457
Normal	ARX(1,1)	EGARCH(1,2)	70.8116	−125.6232	−89.3529
Normal	ARX(1,1)	EGARCH(2,1)	76.9436	−135.8873	−95.0832
Normal	ARX(1,1)	EGARCH(2,2)	76.9436	−133.8873	−88.5494
Normal	ARX(2,1)	GARCH(1,1)	76.4837	−138.9675	−107.2310
Normal	ARX(2,1)	GARCH(1,2)	76.4837	−136.9675	−100.6972
Normal	ARX(2,1)	GARCH(2,1)	76.4848	−136.9695	−100.6992
Normal	ARX(2,1)	GARCH(2,2)	76.5935	−135.1870	−94.3829
Normal	ARX(2,1)	GJR-GARCH(1,1)	77.6015	−139.2029	−102.9326
Normal	ARX(2,1)	GJR-GARCH(1,2)	77.6015	−137.2029	−96.3988
Normal	ARX(2,1)	GJR-GARCH(2,1)	80.0966	−140.1932	−94.8553
Normal	ARX(2,1)	GJR-GARCH(2,2)	82.5502	−143.1004	−93.2288
Normal	ARX(2,1)	EGARCH(1,1)	70.8342	−125.6684	−89.3981
Normal	ARX(2,1)	EGARCH(1,2)	71.0550	−124.1099	−83.3058
Normal	ARX(2,1)	EGARCH(2,1)	77.3960	−134.7920	−89.4541
Normal	ARX(2,1)	EGARCH(2,2)	77.3960	−132.7920	−82.9203
Normal	ARX(2,2)	GARCH(1,1)	76.6971	−137.3941	−101.1238
Normal	ARX(2,2)	GARCH(1,2)	76.6971	−135.3941	−94.5900
Normal	ARX(2,2)	GARCH(2,1)	76.7001	−135.4003	−94.5962
Normal	ARX(2,2)	GARCH(2,2)	76.8234	−133.6467	−88.3088
Normal	ARX(2,2)	GJR-GARCH(1,1)	77.8171	−137.6342	−96.8301
Normal	ARX(2,2)	GJR-GARCH(1,2)	77.8171	−135.6342	−90.2963
Normal	ARX(2,2)	GJR-GARCH(2,1)	80.5584	−139.1168	−89.2451
Normal	ARX(2,2)	GJR-GARCH(2,2)	83.1729	−142.3459	−87.9404
Normal	ARX(2,2)	EGARCH(1,1)	71.0135	−124.0269	−83.2228
Normal	ARX(2,2)	EGARCH(1,2)	71.2294	−122.4588	−77.1209

Distribution	Mean Process	Volatility Process	Best LogLik	AIC	BIC
Normal	ARX(2,2)	EGARCH(2,1)	77.8360	−133.6719	−83.8003
Normal	ARX(2,2)	EGARCH(2,2)	77.8360	−131.6719	−77.2665
Student's t	Constant	GARCH(1,1)	38.7813	−69.5626	−51.4274
Student's t	Constant	GARCH(1,2)	39.7888	−69.5775	−46.9086
Student's t	Constant	GARCH(2,1)	38.7813	−67.5626	−44.8937
Student's t	Constant	GARCH(2,2)	39.7888	−67.5775	−40.3748
Student's t	Constant	GJR-GARCH(1,1)	39.7463	−69.4925	−46.8236
Student's t	Constant	GJR-GARCH(1,2)	40.9705	−69.9410	−42.7383
Student's t	Constant	GJR-GARCH(2,1)	39.7996	−65.5991	−33.8626
Student's t	Constant	GJR-GARCH(2,2)	42.0009	−68.0017	−31.7314
Student's t	Constant	EGARCH(1,1)	36.5166	−63.0332	−40.3642
Student's t	Constant	EGARCH(1,2)	38.1862	−64.3725	−37.1697
Student's t	Constant	EGARCH(2,1)	42.9506	−71.9012	−40.1647
Student's t	Constant	EGARCH(2,2)	42.9506	−69.9012	−33.6309
Student's t	ARX(1,1)	GARCH(1,1)	100.1544	−186.3088	−154.5723
Student's t	ARX(1,1)	GARCH(1,2)	100.1544	−184.3088	−148.0385
Student's t	ARX(1,1)	GARCH(2,1)	100.1553	−184.3106	−148.0403
Student's t	ARX(1,1)	GARCH(2,2)	100.1623	−182.3247	−141.5206
Student's t	ARX(1,1)	GJR-GARCH(1,1)	102.1427	−188.2854	−152.0151
Student's t	ARX(1,1)	GJR-GARCH(1,2)	102.1431	−186.2861	−145.4820
Student's t	ARX(1,1)	GJR-GARCH(2,1)	102.5336	−185.0673	−139.7294
Student's t	ARX(1,1)	GJR-GARCH(2,2)	103.3727	−184.7455	−134.8738
Student's t	ARX(1,1)	EGARCH(1,1)	98.4157	−180.8314	−144.5611
Student's t	ARX(1,1)	EGARCH(1,2)	98.5319	−179.0639	−138.2598
Student's t	ARX(1,1)	EGARCH(2,1)	100.0980	−180.1960	−134.8581
Student's t	ARX(1,1)	EGARCH(2,2)	101.1880	−180.3761	−130.5044
Student's t	ARX(2,1)	GARCH(1,1)	100.2634	−184.5268	−148.2564
Student's t	ARX(2,1)	GARCH(1,2)	100.2634	−182.5268	−141.7227
Student's t	ARX(2,1)	GARCH(2,1)	100.2639	−182.5278	−141.7237
Student's t	ARX(2,1)	GARCH(2,2)	100.2702	−180.5404	−135.2025
Student's t	ARX(2,1)	GJR-GARCH(1,1)	102.4946	−186.9891	−146.1850
Student's t	ARX(2,1)	GJR-GARCH(1,2)	102.4953	−184.9906	−139.6527
Student's t	ARX(2,1)	GJR-GARCH(2,1)	102.9181	−183.8363	−133.9646
Student's t	ARX(2,1)	GJR-GARCH(2,2)	103.8765	−183.7530	−129.3475
Student's t	ARX(2,1)	EGARCH(1,1)	98.7448	−179.4897	−138.6856

Distribution	Mean Process	Volatility Process	Best LogLik	AIC	BIC
Student's t	ARX(2,1)	EGARCH(1,2)	98.8712	−177.7423	−132.4045
Student's t	ARX(2,1)	EGARCH(2,1)	100.6370	−179.2741	−129.4024
Student's t	ARX(2,1)	EGARCH(2,2)	101.6258	−179.2516	−124.8461
Student's t	ARX(2,2)	GARCH(1,1)	100.6088	−183.2175	−142.4134
Student's t	ARX(2,2)	GARCH(1,2)	100.6088	−181.2175	−135.8796
Student's t	ARX(2,2)	GARCH(2,1)	100.6112	−181.2224	−135.8845
Student's t	ARX(2,2)	GARCH(2,2)	100.6194	−179.2387	−129.3670
Student's t	ARX(2,2)	GJR-GARCH(1,1)	102.8861	−185.7722	−140.4343
Student's t	ARX(2,2)	GJR-GARCH(1,2)	102.8861	−183.7722	−133.9005
Student's t	ARX(2,2)	GJR-GARCH(2,1)	103.4125	−182.8251	−128.4196
Student's t	ARX(2,2)	GJR-GARCH(2,2)	104.4932	−182.9864	−124.0471
Student's t	ARX(2,2)	EGARCH(1,1)	99.1348	−178.2695	−132.9316
Student's t	ARX(2,2)	EGARCH(1,2)	99.2458	−176.4916	−126.6200
Student's t	ARX(2,2)	EGARCH(2,1)	101.1255	−178.2510	−123.8455
Student's t	ARX(2,2)	EGARCH(2,2)	102.2040	−178.4080	−119.4687
Gaussian mixture	Constant	GARCH(1,1)	40.8067	−69.6135	−42.4107
Gaussian mixture	Constant	GARCH(1,2)	42.2087	−70.4173	−38.6808
Gaussian mixture	Constant	GARCH(2,1)	40.8067	−67.6135	−35.8770
Gaussian mixture	Constant	GARCH(2,2)	42.2087	−68.4173	−32.1470
Gaussian mixture	Constant	GJR-GARCH(1,1)	43.3968	−72.7935	−41.0570
Gaussian mixture	Constant	GJR-GARCH(1,2)	44.9503	−73.9006	−37.6303
Gaussian mixture	Constant	GJR-GARCH(2,1)	43.3968	−68.7935	−27.9894
Gaussian mixture	Constant	GJR-GARCH(2,2)	45.0894	−70.1788	−24.8409
Gaussian mixture	Constant	EGARCH(1,1)	41.4377	−68.8754	−37.1389
Gaussian mixture	Constant	EGARCH(1,2)	43.0855	−70.1710	−33.9007
Gaussian mixture	Constant	EGARCH(2,1)	45.6972	−73.3943	−32.5902
Gaussian mixture	Constant	EGARCH(2,2)	45.6972	−71.3943	−26.0565
Gaussian mixture	ARX(1,1)	GARCH(1,1)	101.6502	−185.3003	−144.4962
Gaussian mixture	ARX(1,1)	GARCH(1,2)	101.6503	−183.3005	−137.9626
Gaussian mixture	ARX(1,1)	GARCH(2,1)	101.6502	−183.3003	−137.9624
Gaussian mixture	ARX(1,1)	GARCH(2,2)	101.6599	−181.3198	−131.4481
Gaussian mixture	ARX(1,1)	GJR-GARCH(1,1)	103.8056	−187.6112	−142.2734
Gaussian mixture	ARX(1,1)	GJR-GARCH(1,2)	103.8056	−185.6112	−135.7396
Gaussian mixture	ARX(1,1)	GJR-GARCH(2,1)	104.2851	−184.5702	−130.1647
Gaussian mixture	ARX(1,1)	GJR-GARCH(2,2)	104.9765	−183.9530	−125.0138
Gaussian mixture	ARX(1,1)	EGARCH(1,1)	100.9490	−181.8980	−136.5601

Distribution	Mean Process	Volatility Process	Best LogLik	AIC	BIC
Gaussian mixture	ARX(1,1)	EGARCH(1,2)	100.9582	−179.9164	−130.0447
Gaussian mixture	ARX(1,1)	EGARCH(2,1)	102.4071	−180.8143	−126.4088
Gaussian mixture	ARX(1,1)	EGARCH(2,2)	103.2633	−180.5267	−121.5874
Gaussian mixture	ARX(2,1)	GARCH(1,1)	101.8307	−183.6615	−138.3236
Gaussian mixture	ARX(2,1)	GARCH(1,2)	101.8308	−181.6616	−131.7899
Gaussian mixture	ARX(2,1)	GARCH(2,1)	101.8307	−181.6615	−131.7898
Gaussian mixture	ARX(2,1)	GARCH(2,2)	101.8386	−179.6772	−125.2718
Gaussian mixture	ARX(2,1)	GJR-GARCH(1,1)	104.4273	−186.8546	−136.9829
Gaussian mixture	ARX(2,1)	GJR-GARCH(1,2)	104.4273	−184.8546	−130.4492
Gaussian mixture	ARX(2,1)	GJR-GARCH(2,1)	104.9099	−183.8198	−124.8805
Gaussian mixture	ARX(2,1)	GJR-GARCH(2,2)	105.6055	−183.2110	−119.7380
Gaussian mixture	ARX(2,1)	EGARCH(1,1)	101.6090	−181.2179	−131.3462
Gaussian mixture	ARX(2,1)	EGARCH(1,2)	101.6161	−179.2322	−124.8267
Gaussian mixture	ARX(2,1)	EGARCH(2,1)	103.1390	−180.2781	−121.3388
Gaussian mixture	ARX(2,1)	EGARCH(2,2)	103.8033	−179.6065	−116.1335
Gaussian mixture	ARX(2,2)	GARCH(1,1)	102.0243	−182.0486	−132.1769
Gaussian mixture	ARX(2,2)	GARCH(1,2)	102.0243	−180.0486	−125.6431
Gaussian mixture	ARX(2,2)	GARCH(2,1)	102.0243	−180.0486	−125.6431
Gaussian mixture	ARX(2,2)	GARCH(2,2)	102.0359	−178.0717	−119.1325
Gaussian mixture	ARX(2,2)	GJR-GARCH(1,1)	104.7171	−185.4342	−131.0287
Gaussian mixture	ARX(2,2)	GJR-GARCH(1,2)	104.7171	−183.4342	−124.4949
Gaussian mixture	ARX(2,2)	GJR-GARCH(2,1)	105.2849	−182.5698	−119.0968
Gaussian mixture	ARX(2,2)	GJR-GARCH(2,2)	106.1043	−182.2086	−114.2018
Gaussian mixture	ARX(2,2)	EGARCH(1,1)	101.9001	−179.8001	−125.3947
Gaussian mixture	ARX(2,2)	EGARCH(1,2)	101.9044	−177.8089	−118.8696
Gaussian mixture	ARX(2,2)	EGARCH(2,1)	103.5168	−179.0336	−115.5606
Gaussian mixture	ARX(2,2)	EGARCH(2,2)	104.2678	−178.5357	−110.5288

6.b. *Best Models by Criterion and Volatility Family*

For each criterion and each volatility family, the selected model is the best across mean-process choices, orders, and distributions using stable & successful fits. The main result tables contain 144 reported model combinations.

6.b.i. *AIC:*

GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M061`	Student’s t	ARX(1,1)	GARCH(1,1)	100.1544	−186.3088	−154.5723

Mean process:

$$\hat{\pi}_{t+1} = 0.0148 + 0.4441\,\pi_t + 0.5718\,SPF_t + \mu_{t+1} \tag{43}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0061 + 0.2477\,u_{t-1}^2 + 0.6670\,\sigma_{t-1}^2 \tag{44}$$

Distribution parameters:

Parameter	Estimate
ν	5.0899

Parameter	Coef	Std Err	t-value	p-value
c	0.0148	0.0180	0.8241	0.4099
ρ_1	0.4441	0.0451	9.8407	0.0000
ϕ_1	0.5718	0.0760	7.5191	0.0000
ω	0.0061	0.0024	2.5927	0.0095
α_1	0.2477	0.0666	3.7198	0.0002
β_1	0.6670	0.0735	9.0700	0.0000
ν	5.0899	0.8695	5.8540	0.0000

GJR-GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M062`	Student's t	ARX(1,1)	GJR-GARCH(1,1)	102.1427	−188.2854	−152.0151

Mean process:

$$\hat{\pi}_{t+1} = 0.0216 + 0.4340\,\pi_t + 0.5660\,SPF_t + \mu_{t+1} \tag{45}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0054 + 0.3015\,\left(u_{t-1}^{+}\right)^2 + 0.1484\,\left(u_{t-1}^{-}\right)^2 + 0.6979\,\sigma_{t-1}^2 \tag{46}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

Distribution parameters:

Parameter	Estimate
ν	5.0138

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.0216	0.0183	1.1819	0.2373

Parameter	Coef	Std Err	t-value	p-value
ρ_1	0.4340	0.0440	9.8613	0.0000
ϕ_1	0.5660	0.0741	7.6411	0.0000
ω	0.0054	0.0020	2.7726	0.0056
α_1	0.3015	0.0733	4.1159	0.0000
$\gamma_1 - \alpha_1$	−0.1531	0.0695	−2.2040	0.0275
β_1	0.6979	0.0695	10.0370	0.0000
ν	5.0138	0.8531	5.8772	0.0000

EGARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M111`	Gaussian mixture	ARX(1,1)	EGARCH(1,1)	100.9490	−181.8980	−136.5601

Mean process:

$$\hat{\pi}_{t+1} = 0.0176 + 0.4272\,\pi_t + 0.5975\,SPF_t + \mu_{t+1} \tag{47}$$

Volatility process:

$$\ln \hat{\sigma}_t^2 = -0.2797 + 0.3475\left(\mid z_{t-1} \mid - \sqrt{2/\pi}\right) + 0.0628\,z_{t-1} + 0.9011\ln \sigma_{t-1}^2 \tag{48}$$

Distribution parameters:

Parameter	Estimate
p_1	0.8289
μ_1	0.0039
σ_1^2	0.5544
p_2	0.1711
μ_2	−0.0191
σ_2^2	3.1581

Parameter	Coef	Std Err	t-value	p-value
c	0.0176	0.0165	1.0676	0.2857
ρ_1	0.4272	0.0422	10.1279	0.0000
ϕ_1	0.5975	0.0669	8.9321	0.0000
ω	−0.2797	0.1456	−1.9212	0.0547
α_1	0.3475	0.0960	3.6209	0.0003
γ_1	0.0628	0.0370	1.6985	0.0894
β_1	0.9011	0.0490	18.3830	0.0000

Parameter	Coef	Std Err	t-value	p-value
p_1	0.8289	0.0604	13.7212	0.0000
μ_1	0.0039	0.0405	0.0974	0.9224
σ_1^2	0.5544	0.0690	8.0391	0.0000

6.b.ii. *BIC:*

GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M061`	Student's t	ARX(1,1)	GARCH(1,1)	100.1544	−186.3088	−154.5723

Mean process:

$$\hat{\pi}_{t+1} = 0.0148 + 0.4441\,\pi_t + 0.5718\,SPF_t + \mu_{t+1} \tag{49}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0061 + 0.2477\,u_{t-1}^2 + 0.6670\,\sigma_{t-1}^2 \tag{50}$$

Distribution parameters:

Parameter	Estimate
ν	5.0899

Parameter	Coef	Std Err	t-value	p-value
c	0.0148	0.0180	0.8241	0.4099
ρ_1	0.4441	0.0451	9.8407	0.0000
ϕ_1	0.5718	0.0760	7.5191	0.0000
ω	0.0061	0.0024	2.5927	0.0095
α_1	0.2477	0.0666	3.7198	0.0002
β_1	0.6670	0.0735	9.0700	0.0000
ν	5.0899	0.8695	5.8540	0.0000

GJR-GARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M062`	Student's t	ARX(1,1)	GJR-GARCH(1,1)	102.1427	−188.2854	−152.0151

Mean process:

$$\hat{\pi}_{t+1} = 0.0216 + 0.4340\,\pi_t + 0.5660\,SPF_t + \mu_{t+1} \tag{51}$$

Volatility process:

$$\hat{\sigma}_t^2 = 0.0054 + 0.3015\,\left(u_{t-1}^+\right)^2 + 0.1484\,\left(u_{t-1}^-\right)^2 + 0.6979\,\sigma_{t-1}^2 \tag{52}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

Distribution parameters:

Parameter	Estimate
ν	5.0138

For GJR-GARCH, $\gamma_i - \alpha_i$ is the raw `arch` indicator coefficient; the equation above reports the negative-shock coefficient γ_i.

Parameter	Coef	Std Err	t-value	p-value
c	0.0216	0.0183	1.1819	0.2373
ρ_1	0.4340	0.0440	9.8613	0.0000
ϕ_1	0.5660	0.0741	7.6411	0.0000
ω	0.0054	0.0020	2.7726	0.0056
α_1	0.3015	0.0733	4.1159	0.0000
$\gamma_1 - \alpha_1$	−0.1531	0.0695	−2.2040	0.0275
β_1	0.6979	0.0695	10.0370	0.0000
ν	5.0138	0.8531	5.8772	0.0000

EGARCH:

Model ID	Distribution	Mean process	Volatility process	Log likelihood	AIC	BIC
`M063`	Student's t	ARX(1,1)	EGARCH(1,1)	98.4157	−180.8314	−144.5611

Mean process:

$$\hat{\pi}_{t+1} = 0.0175 + 0.4342\,\pi_t + 0.5867\,SPF_t + \mu_{t+1} \tag{53}$$

Volatility process:

$$\ln \hat{\sigma}_t^2 = -0.3089 + 0.3702\left(|\,z_{t-1}\,| - \sqrt{2/\pi}\right) + 0.0551\,z_{t-1} + 0.8902\ln \sigma_{t-1}^2 \tag{54}$$

Distribution parameters:

Parameter	Estimate
ν	4.7812

Parameter	Coef	Std Err	t-value	p-value
c	0.0175	0.0044	3.9949	0.0001
ρ_1	0.4342	0.0233	18.6127	0.0000
ϕ_1	0.5867	0.0353	16.6237	0.0000
ω	−0.3089	0.1518	−2.0349	0.0419
α_1	0.3702	0.1193	3.1018	0.0019

γ_1	0.0551	0.0395	1.3949	0.1631
Parameter	Coef	Std Err	t-value	p-value
β_1	0.8902	0.0520	17.1277	0.0000
ν	4.7812	0.7685	6.2216	0.0000

7. Regime-Switching Inflation Models

Let the latent state be

$$s_t \in \{1, ..., K\}, \qquad \Pr(s_t = j \mid s_{t-1} = i) = p_{ij}. \tag{55}$$

I estimate several specifications that differ in **which blocks of parameters are allowed to switch with s_t**. Taking the ARX(2,2) model as an example,

$$\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 \, \mathrm{SPF}_t + \phi_2 \, \mathrm{SPF}_{t-1} + \mu_{t+1}, \tag{56}$$

the parameters are grouped into three blocks, and each block can independently be made regime-dependent:

Block	State-dependent parameters
AR component	$c_{s_t}, \ \rho_{1,s_t}, \ \rho_{2,s_t}$
SPF component	$\phi_{1,s_t}, \ \phi_{2,s_t}$
Shock distribution	σ_{s_t}

A given specification is defined by selecting any subset of these three blocks to be switching; parameters in blocks not selected are held constant across regimes.

7.a. Estimation And Collection

For each regime-switching specification, I run 50 optimizer starts. Each start first uses 50 EM iterations to improve the starting values and then applies MLE to fine tune the likelihood. The collected estimate for a model specification is the highest log-likelihood attempt that passes all checks:

- optimizer convergence,
- AR mean-process stationarity in every regime,
- positive non-boundary variance,
- and valid non-boundary transition probabilities.

The collection step rejects selected attempts with variance estimates at or below `max(1e-6, 1e-3 * var(y))`, where `y` is the dependent variable of the specification (`u_t` for the Constant mean process, `\pi_t` for the ARX mean processes), or transition probabilities within `1e-4` of `0` or `1`.

For the AR stationarity check, only the inflation-lag coefficients enter the AR polynomial. SPF coefficients are treated as exogenous loadings. Specifications where none of the 50 attempts passes all checks are excluded from the results table; they are not reported even as a fallback, because the highest-likelihood attempt for such specifications is typically a degenerate solution (e.g., a collapsed regime variance that inflates the log-likelihood artificially).

By default only normal-residual specifications are estimated. Student's t variants can be estimated as a robustness exercise with the `--include-student-t` flag, but they are always excluded from the reported tables.

7.b. Data Frequency

By default `estimate_regime_switching_models.py` estimates on the quarterly effective sample (`DataSummary/Aggregate_CPI_inflation_Quarterly.pkl`, 1969Q2–2026Q1, 228 observations after the 2026-07-14 data update; the committed quarterly results in `results/` were estimated on the previous

1969Q2–2022Q4, 215-observation sample) and writes to `results/`. The same specification menu can be estimated on the monthly effective sample (`DataSummary/Aggregate_CPI_inflation_Monthly.pkl`, 1969M2–2026M5, 688 observations) with the `--data-path` and `--output-dir` flags; monthly results are stored in `results_monthly/`:

```
python estimate_regime_switching_models.py \
    --data-path ../DataSummary/Aggregate_CPI_inflation_Monthly.pkl \
    --output-dir results_monthly
```

All model settings (specification menu, starts, checks, and selection policy) are identical across the two frequencies.

Each model is selected from 50 starts; only estimates that passed all checks (optimizer convergence, AR stationarity, parameter bounds, and valid transition probabilities) are reported. Specifications where no start passed all checks are excluded. `Sw.Dist` reflects whether σ^2 is regime-specific.

Mean	K	Sw.AR	Sw.SPF	Sw.Dist	LogLik	AIC	BIC
Constant	2	N	N	Y	−186.6300	381.2600	394.7425
Constant	3	N	N	Y	−180.6569	379.3139	409.6496
ARX(1,1)	2	N	N	Y	−177.1889	368.3779	391.9723
ARX(1,1)	2	Y	N	Y	−175.1175	368.2350	398.5708
ARX(1,1)	2	Y	Y	Y	−175.0924	370.1848	403.8912
ARX(1,1)	3	N	N	Y	−174.3093	372.6186	413.0662
ARX(1,1)	3	Y	N	Y	−162.3751	356.7503	410.6805
ARX(2,1)	2	N	N	Y	−176.2156	368.4311	395.3962
ARX(2,1)	2	Y	N	Y	−174.9422	371.8845	408.9615
ARX(2,1)	2	Y	Y	Y	−174.3768	372.7535	413.2012
ARX(2,1)	3	Y	N	Y	−157.6865	353.3729	417.4150
ARX(2,2)	2	N	N	Y	−175.9610	369.9221	400.2578
ARX(2,2)	2	Y	N	Y	−174.5337	373.0674	413.5150
ARX(2,2)	2	Y	Y	Y	−174.1552	376.3105	423.4994

8. Best Regime-Switching Model

Selected by AIC among estimates that passed optimizer convergence, stationarity, parameter-bound, distribution, and transition checks.

8.a. *Model Summary*

Mean process	Regime-switching parts	Non-switching parts	LogLik	AIC	BIC
ARX(2,1)	AR block, including c, σ^2	SPF block	−157.6865	353.3729	417.4150

8.b. *Switching Blocks*

Block	Switching
AR block, including c	Y
SPF block	N
Shock distribution (σ^2)	Y

8.c. *Mean Process*

$$\pi_{t+1} = c_{s_t} + \rho_{1,s_t}\,\pi_t + \rho_{2,s_t}\,\pi_{t-1} + \phi_1\,SPF_t + u_{t+1} \tag{57}$$

8.d. *Mean Parameter Values*

Parameter	Switching	Regime 0	Regime 1	Regime 2
c	Y	0.2893	−0.1234	0.1561
ρ_1	Y	0.0142	0.1989	0.2226
ρ_2	Y	−0.1378	0.4114	−0.0895
ϕ_1	N	0.7487	0.7487	0.7487

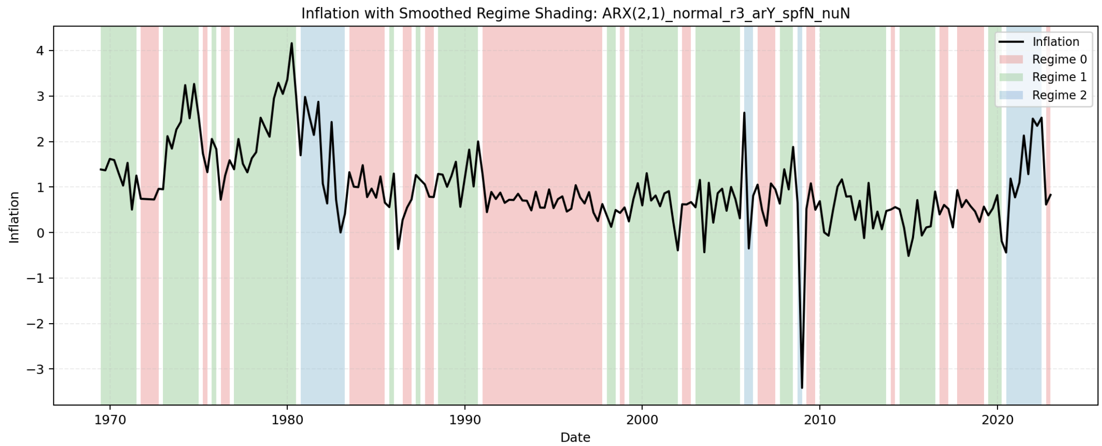
8.e. *Distribution Parameters*

Parameter	Switching	Regime 0	Regime 1	Regime 2
σ^2	Y	0.0517	0.1973	1.4556

8.f. *Transition Probability Matrix*

From \ To	Regime 0	Regime 1	Regime 2	Row Sum
Regime 0	0.7705	0.2291	0.0004	1.0000
Regime 1	0.1010	0.8367	0.0623	1.0000
Regime 2	0.2227	0.0005	0.7768	1.0000

8.g. *Regime Classification Plot*


Inflation with Smoothed Regime Shading: ARX(2,1)_normal_r3_arY_spfN_nuN
Inflation
Regime 0
Regime 1
Regime 2
4
3
2
1
0
−1
−2
−3
1970
1980
1990
2000
2010
2020
Date

Each model is selected from 50 starts; only estimates that passed all checks (optimizer convergence, AR stationarity, parameter bounds, and valid transition probabilities) are reported. Specifications where no start passed all checks are excluded. `Sw.Dist` reflects whether σ^2 is regime-specific.

Mean	K	Sw.AR	Sw.SPF	Sw.Dist	LogLik	AIC	BIC
Constant	2	N	N	Y	21.0036	−34.0071	−15.8720
ARX(1,1)	2	N	N	Y	91.9010	−169.8020	−138.0655
ARX(1,1)	2	Y	N	Y	93.7691	−169.5382	−128.7341
ARX(1,1)	2	Y	Y	Y	93.7702	−167.5405	−122.2026
ARX(1,1)	3	N	N	Y	91.9024	−159.8047	−105.3992
ARX(1,1)	3	Y	N	Y	112.5283	−193.0566	−120.5160
ARX(2,1)	2	N	N	Y	91.9497	−167.8995	−131.6292
ARX(2,1)	2	Y	N	Y	94.1403	−166.2805	−116.4089
ARX(2,1)	2	Y	Y	Y	94.1451	−164.2902	−109.8847
ARX(2,1)	3	N	N	Y	91.9657	−157.9314	−98.9921
ARX(2,1)	3	Y	N	Y	119.1499	−200.2998	−114.1578
ARX(2,2)	2	N	N	Y	92.0116	−166.0232	−125.2191
ARX(2,2)	2	Y	N	Y	94.2497	−164.4994	−110.0939
ARX(2,2)	2	Y	Y	Y	95.0807	−162.1613	−98.6883
ARX(2,2)	3	N	N	Y	102.9279	−177.8558	−114.3828
ARX(2,2)	3	Y	N	Y	119.9819	−199.9639	−109.2881

9. Best Regime-Switching Model

Selected by AIC among estimates that passed optimizer convergence, stationarity, parameter-bound, distribution, and transition checks.

9.a. Model Summary

Mean process	Regime-switching parts	Non-switching parts	LogLik	AIC	BIC
ARX(2,1)	AR block, including c, σ^2	SPF block	119.1499	−200.2998	−114.1578

9.b. Switching Blocks

Block	Switching
AR block, including c	Y
SPF block	N
Shock distribution (σ^2)	Y

9.c. Mean Process

$$\pi_{t+1} = c_{s_t} + \rho_{1,s_t}\,\pi_t + \rho_{2,s_t}\,\pi_{t-1} + \phi_1\,SPF_t + u_{t+1} \tag{58}$$

9.d. Mean Parameter Values

Parameter	Switching	Regime 0	Regime 1	Regime 2
c	Y	0.2179	−0.0112	0.0480
ρ_1	Y	0.0504	0.5104	0.5410
ρ_2	Y	−0.3566	0.1595	−0.0440
ϕ_1	N	0.4716	0.4716	0.4716

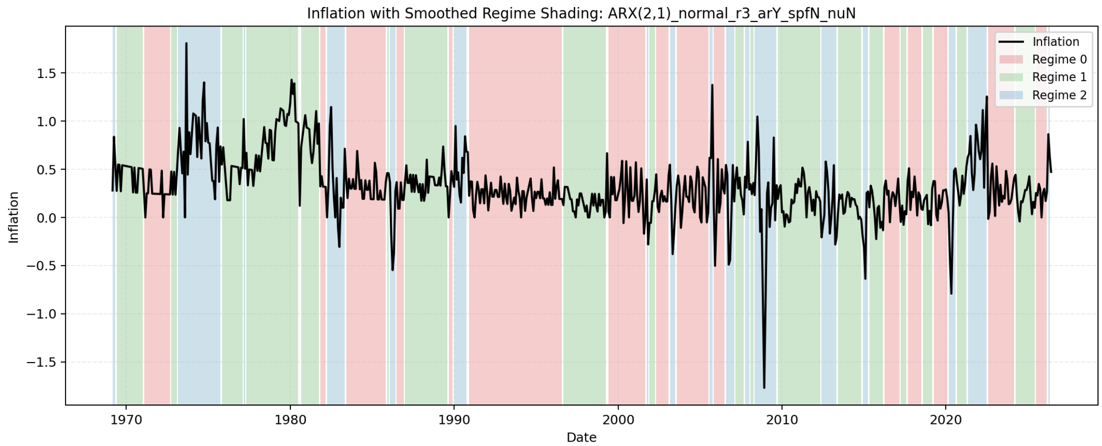
9.e. Distribution Parameters

Parameter	Switching	Regime 0	Regime 1	Regime 2
σ^2	Y	0.0174	0.0197	0.1595

9.f. Transition Probability Matrix

From \ To	Regime 0	Regime 1	Regime 2	Row Sum
Regime 0	0.9142	0.0439	0.0419	1.0000
Regime 1	0.0399	0.9073	0.0528	1.0000
Regime 2	0.0598	0.0973	0.8429	1.0000

9.g. Regime Classification Plot


Inflation with Smoothed Regime Shading: ARX(2,1)_normal_r3_arY_spfN_nuN
Inflation
Regime 0
Regime 1
Regime 2
Inflation
Date
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
1970
1980
1990
2000
2010
2020

10. BEGE GARCH Family

All BEGE specifications use the same effective sample, residual definition, mean-process menu, random-search protocol, density evaluator, and result selection rules. The model-specific shape recursions and restrictions are reported.

10.a. Mean Processes

The BEGE exercises estimate all four mean processes:

- **Constant**: $\pi_{t+1} = SPF_t + u_{t+1}$.
- **ARX(1,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + u_{t+1}$.
- **ARX(2,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + u_{t+1}$.
- **ARX(2,2)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + u_{t+1}$.

The mean-parameter bounds used in estimation and collection are:

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	–	–	–	–	–
ARX(1,1)	$(\min \pi_t, \max \pi_t)$	$[-1, 1]$	–	$[-10, 10]$	–
ARX(2,1)	$(\min \pi_t, \max \pi_t)$	$[-2, 2]$	$[-1, 1]$	$[-10, 10]$	–
ARX(2,2)	$(\min \pi_t, \max \pi_t)$	$[-2, 2]$	$[-1, 1]$	$[-10, 10]$	$[-10, 10]$

Best-model collection also imposes mean-process stationarity before a BEGE estimate is eligible for reporting. For ARX(1,1), this requires $\mid \rho_1 \mid < 1$. For ARX(2,1) and ARX(2,2), the stationarity condition is the usual AR(2) polynomial restriction:

$$1 - \rho_1 z - \rho_2 z^2 = 0 \tag{59}$$

must have both roots outside the unit circle. Equivalently, the reported estimate must satisfy

$$\rho_1 + \rho_2 < 1, \qquad \rho_2 - \rho_1 < 1, \qquad \rho_2 > -1. \tag{60}$$

Only the inflation-lag coefficients ρ_1 and ρ_2 enter this stationarity check; the SPF coefficients ϕ_1 and ϕ_2 are treated as exogenous-regressor loadings and are not included in the AR polynomial.

Starting values for the estimated mean parameters are drawn uniformly from OLS-centered intervals, $\hat{\theta}_j \pm 2\, SE\big(\hat{\theta}_j\big)$.

10.b. Residual Dynamics

Each BEGE model writes the inflation residual as

$$u_t = \sigma_p \omega_{p,t} - \sigma_n \omega_{n,t}, \qquad \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \quad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1), \tag{61}$$

where $\tilde{\Gamma}(a, 1)$ is a centered gamma shock with shape a and unit scale. Conditional on p_t and n_t,

- conditional variance: $\sigma_p^2 p_t + \sigma_n^2 n_t$,
- conditional skewness numerator: $2\big(\sigma_p^3 p_t - \sigma_n^3 n_t\big)$,
- conditional excess-kurtosis numerator: $6\big(\sigma_p^4 p_t + \sigma_n^4 n_t\big)$.

10.c. BEGE Volatility Specifications

The common parameter bounds used in estimation and result collection are:

Parameter group	Bound
Fixed shapes $\bar{p}, \bar{n}$ and intercepts p_0, n_0	$[0, 10]$
Persistence parameters ρ, ρ_p, ρ_n	$[0, 1]$
Shock-loading parameters $\phi^{\pm}, \phi_p, \phi_n, \phi_p^{\pm}, \phi_n^{\pm}$	$[0, 2]$
Scale parameters σ_p, σ_n	$[10^{-5}, 2]$

To rule out numerically degenerate and explosive variance paths, I use an EWMA path as a scale-adaptive reference for the BEGE implied variance. This is a loose numerical safeguard, not a variance-targeting restriction.[4] For residuals u_t, define

$$EWMA_1 = \frac{\sum_{j=1}^{\tau} \lambda^{j-1} u_j^2}{\sum_{j=1}^{\tau} \lambda^{j-1}}, \qquad EWMA_t = \lambda EWMA_{t-1} + (1-\lambda) u_{t-1}^2, \quad t \geq 2. \tag{62}$$

The EWMA implied-variance bounds are imposed during optimization and checked again during collection. With $\lambda = 0.94$ and $\tau = \min(75, T)$, let $s_u^2 := 1/T \sum_{j=1}^{T} u_j^2$ and $M_u := 1 + \max_{1 \leq s \leq T} u_s^2$. Then

$$\begin{aligned} \underline{v}_t &:= \max\{10^{-6} EWMA_t,\ 10^{-8} s_u^2\}, \\ \overline{v}_t &:= \max\{\min(10^6 EWMA_t,\ 10^7 M_u),\ M_u\}, \\ \underline{v}_t &\leq \sigma_p^2 p_t + \sigma_n^2 n_t \leq \overline{v}_t, \qquad t = 1, \dots, T. \end{aligned} \tag{63}$$

10.c.i. *Constant BEGE:*

The Constant BEGE model keeps both shape parameters fixed:

$$p_t = \bar{p}, \qquad n_t = \bar{n}. \tag{64}$$

Thus $u_t = \sigma_p \omega_{p,t} - \sigma_n \omega_{n,t}$ with $\omega_{p,t} \sim \tilde{\Gamma}(\bar{p}, 1)$ and $\omega_{n,t} \sim \tilde{\Gamma}(\bar{n}, 1)$.

10.c.ii. *Symmetric BEGE:*

The Symmetric BEGE model uses the same persistence and shock-loading parameters in both shape recursions:

$$\begin{aligned} p_t &= p_0 + \rho p_{t-1} + \frac{\phi^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_p^2} \left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho n_{t-1} + \frac{\phi^+}{2\sigma_n^2} \left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2. \end{aligned} \tag{65}$$

The stability restriction is

$$\rho + \frac{\phi^+ + \phi^-}{2} < 1. \tag{66}$$

[4] The exponential weights follow the variance-forecasting idea in J.P. Morgan/Reuters, *RiskMetrics—Technical Document*, Fourth Edition (1996), Chapter 5, where recent squared shocks receive more weight than older shocks. Here the EWMA path is only a smoothed proxy for the local scale of u_t^2, not a BEGE variance target. The loose envelope is adapted from the `arch` package's `VolatilityProcess.variance_bounds`: the lower bound prevents the implied BEGE variance from collapsing toward zero, while the upper bound screens out explosive variance paths that would make likelihood evaluation numerically unreliable.

10.c.iii. *BadGood BEGE:*

The BadGood BEGE model lets each shape process react symmetrically to squared residuals, with separate good- and bad-environment parameters:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p}{2\sigma_p^2} u_{t-1}^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n}{2\sigma_n^2} u_{t-1}^2. \end{aligned} \tag{67}$$

The stability restrictions are

$$\rho_p + \phi_p < 1, \qquad \rho_n + \phi_n < 1. \tag{68}$$

10.c.iv. *Inflation/Deflation BEGE:*

The Inflation/Deflation BEGE model lets the good-news shape respond only to positive residuals and the bad-news shape respond only to negative residuals:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2. \end{aligned} \tag{69}$$

The stability restrictions are

$$\rho_p + \frac{\phi_p^+}{2} < 1, \qquad \rho_n + \frac{\phi_n^-}{2} < 1. \tag{70}$$

10.c.v. *Constant-p Full BEGE:*

Constant-p Full BEGE fixes the good-news shape and keeps the unrestricted Full BEGE update for the bad-news shape:

$$\begin{aligned} p_t &= p_0, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2} \left(u_{t-1}^+\right)^2 + \frac{\phi_n^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2. \end{aligned} \tag{71}$$

The dynamic shape process satisfies

$$\rho_n + \frac{\phi_n^+ + \phi_n^-}{2} < 1. \tag{72}$$

10.c.vi. *Constant-n Full BEGE:*

Constant-n Full BEGE fixes the bad-news shape and keeps the unrestricted Full BEGE update for the good-news shape:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2 + \frac{\phi_p^-}{2\sigma_p^2} \left(u_{t-1}^-\right)^2, \\ n_t &= n_0. \end{aligned} \tag{73}$$

The dynamic shape process satisfies

$$\rho_p + \frac{\phi_p^+ + \phi_p^-}{2} < 1. \tag{74}$$

10.c.vii. *Full BEGE:*

The Full BEGE model allows each shape process to respond separately to positive and negative residuals:

$$\begin{aligned} p_t &= p_0 + \rho_p p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2}\left(u_{t-1}^+\right)^2 + \frac{\phi_p^-}{2\sigma_p^2}\left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho_n n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2}\left(u_{t-1}^+\right)^2 + \frac{\phi_n^-}{2\sigma_n^2}\left(u_{t-1}^-\right)^2. \end{aligned} \tag{75}$$

The stability restrictions are

$$\rho_p + \frac{\phi_p^+ + \phi_p^-}{2} < 1, \qquad \rho_n + \frac{\phi_n^+ + \phi_n^-}{2} < 1. \tag{76}$$

10.d. Random Search

BEGE estimation uses 50 independent seed jobs for each volatility specification. Within each seed job, each mean process uses 40 saved random draws, and each draw runs 25 optimizer starts. Therefore each BEGE specification and mean-process pair uses

$$50 \times 40 \times 25 = 50,000 \tag{77}$$

optimizer starts.

For volatility parameters, starting values are sampled inside the documented bounds. Shape intercept starts p_0, n_0 are sampled from the positive part of $[0, 10]$, scale starts σ_p, σ_n from $\left[10^{-5}, 2\right]$, and persistence and shock-loading starts are drawn so the relevant stability restriction is satisfied before optimization begins.

When a dynamic BEGE recursion is evaluated, I initialize the pre-sample p or n state by the parameter-implied unconditional backcast, $p_0/\left(1 - \rho - \frac{1}{2}(\phi^+ + \phi^-)\right)$ or the corresponding restricted formula, with a small positive numerical floor.[5] Constant-p and Constant-n models set the constant shape path directly.

10.e. Density Evaluation

Likelihood evaluation uses the stabilized BEGE density implementation documented in the BEGE Density section. The current evaluator keeps the closed-form hypergeometric expression in stable regions, switches to a guarded saddlepoint approximation when the closed form becomes numerically fragile.

10.f. Selection And Reporting

For each BEGE specification, the reported best-model page contains only the likelihood-best admissible estimate across mean processes. The by-mean best rows remain available in the CSV outputs linked from each report. Selection requires successful optimizer convergence, mean-process stationarity, documented parameter and stability constraints, and EWMA implied-variance bounds.

[5] The numerical floor is 10^{-4}: if the initial backcast or a later recursion produces a smaller value p_t, n_t, the code uses 0.0001 instead.

Each report also gives empirical 5%, median, and 95% quantiles for p_t, n_t, $\mathrm{Var}_t(u_t)$, the skewness numerator, and the excess-kurtosis numerator.

10.g. Standard Errors

I compute standard errors for the reported best estimate. For each observation, I approximate the score —the first derivative of that observation's log likelihood with respect to the parameters—using centered finite differences. I approximate the Hessian—the matrix of second derivatives of the total negative log likelihood—using the four-point centered finite-difference method implemented by `statsmodels.tools.numdiff.approx_hess`.

The default covariance is the sandwich matrix

$$\hat{V}\left(\hat{\theta}\right) = \hat{H}^{-1}\left(\sum_{t=1}^{T} \hat{g}_t {\hat{g}_t}'\right)\hat{H}^{-1}, \tag{78}$$

where $\hat{H}$ is the Hessian of the negative log likelihood and $\hat{g}_t$ is the numerical score contribution for observation t. If the sandwich calculation is numerically unstable, the collector uses an observed-information or inverse-OPG fallback. Standard errors that cannot be reliably identified are reported as `NA`.

11. BEGE Density Evaluation

The BEGE (Bad Environment–Good Environment) shock is defined as the difference of two centered gamma variables,

$$u_t = \sigma_p \left(\Gamma(p_t, 1) - p_t \right) - \sigma_n \left(\Gamma(n_t, 1) - n_t \right), \tag{79}$$

where p_t and n_t are the time-varying shape parameters driven by the GARCH recursion and σ_p, σ_n are fixed scale parameters. The density $f\left(u_t \mid p_t, n_t, \sigma_p, \sigma_n\right)$ enters the log-likelihood at every observation, so both accuracy and speed of evaluation matter for estimation.

This section compares three implementations of the BEGE log density:

- **Justin's closed-form** implementation, which evaluates the exact analytic expression involving the confluent hypergeometric function U;
- the **stabilized** implementation (current), which guards Justin's closed-form expression with a saddlepoint approximation in fragile large-shape regions;
- **convolution-based numerical integration**, which is slow but serves as an independent benchmark.

The main finding is that the closed-form formula is accurate for small and moderate shape values, but becomes numerically fragile when the recursive shapes enter a transitional range where the BEGE distribution is already close to Gaussian. In that range, large log terms in the hypergeometric expression nearly cancel, and finite-precision arithmetic can produce artificial pointwise log densities that are orders of magnitude too large, which can dominate the likelihood search.

11.a. Executive Summary

- For moderate shape values, the stabilized and closed-form implementations agree to numerical precision.
- On the fixed-shape timing tests below, the stabilized implementation is about 1.5 times faster than the closed-form implementation and about 957 times faster than 5,000-grid numerical integration at the median across the five test cases.
- Stored estimates from earlier production runs exhibited implausibly large likelihoods in a subset of dynamic BEGE specifications. For example, one ARX(2,1) BadGood row had a stored log-likelihood of 4920.7089, whereas the stabilized implementation gives −722.4342 and the saddlepoint approximation gives −722.7255.
- At the pointwise level, one anomalous row reported $\ell_t = 90.3648$ for an observation with conditional variance 42.9373. The stabilized, saddlepoint, and Fourier-inversion values all give approximately −2.8008. This confirms a numerical failure of the hypergeometric evaluation, not an economic feature of the BEGE dynamics.

11.b. Implementation Design

The table below summarizes the design choices in the stabilized implementation relative to Justin's closed-form implementation.

Issue	Justin's closed-form	Stabilized implementation	Rationale

Hypergeometric evaluation	Direct `scipy.special.hyperu` with scalar `mpmath.hyperu` fallback.	Vectorized SciPy where numerically stable; selective high-precision `mpmath` fallback; log-domain asymptotic fallback when `hyperu` is nonfinite.	Avoids the per-observation cost of high-precision arithmetic while preserving stable exact values.
Large-shape transition region	Closed-form branch until a hard maximum-shape cutoff.	Guards exact values once $p_t + n_t \geq 40$; smooth log-sum-exp blend between total shape 50 and 80; replaces exact values when exact and saddlepoint disagree by more than 2 log units.	Numerical failures are driven by cancellation in total shape and can occur before a single-shape cutoff is reached.
Near-normal limit	No explicit Gaussian-limit rule.	Substitutes a Gaussian density when total shape exceeds 500 and standardized skewness and excess kurtosis are both below 0.03.	Gamma shocks converge analytically to a normal law under variance-preserving shape rescaling.
Density continuity	Hard branch switches can create likelihood discontinuities.	Log-sum-exp blending in the transition band.	Keeps the likelihood smoother for numerical optimization.
Failure diagnostics	High likelihoods can enter stored results silently.	The collector recomputes stored estimates with the stabilized density and writes per-path and per-layer diagnostics.	Prevents stale or unstable density values from determining best models.

The guarding thresholds are deliberately conservative: exact hypergeometric values are retained when they agree with the saddlepoint approximation, and large-shape exact values are replaced by the saddlepoint approximation only when the two disagree beyond the stated tolerance.

The accuracy and speed comparisons below are performed on the ARX(1,1) residuals from the canonical effective sample (215 observations). As a reference, the Gaussian OLS log-likelihood is −207.381. The residuals are mean-zero with a standard deviation of 0.636326, skewness of −1.047441, and excess kurtosis of 8.210174.

11.c. Shape Parameter Range

The ranges below use eligible ARX(1,1) rows from the BadGood BEGE results. For each stored parameter vector the recursive shape paths are recomputed on the common ARX(1,1) residuals; the density comparison holds these shape values constant across time.

Model	Rows	p q05	p median	p q95	n q05	n median	n q95	σ_p median	σ_n median
BadGood	994	1.4285	2.8130	6.4330	0.0714	0.3320	2.8342	0.2355	0.5555

11.d. Fixed-Shape Parameter Sets

The log-likelihood and timing comparison uses five representative parameter sets drawn from the BadGood quantiles. In each case σ_p and σ_n are held at their median values.

Set	p	n	σ_p	σ_n
p at q05, n at median	1.428538	0.331970	0.235491	0.555532
p at median, n at median	2.813004	0.331970	0.235491	0.555532
p at q95, n at median	6.433015	0.331970	0.235491	0.555532
p at median, n at q05	2.813004	0.071355	0.235491	0.555532
p at median, n at q95	2.813004	2.834194	0.235491	0.555532

11.e. Log-Likelihood Comparison

The stabilized and Justin’s closed-form implementations are compared against convolution-based numerical integration at several grid sizes.

Parameter set	Stabilized	Justin’s closed-form	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p q05, n median	−214.162839	−214.162839	−223.900355	−213.330306	−214.116656	−214.148282	−214.146613	−214.160080
p median, n median	−188.957117	−188.957117	−196.801762	−188.438308	−189.359597	−189.033173	−188.947127	−188.954400
p q95, n median	−194.340481	−194.340481	−221.076698	−196.996227	−192.679189	−194.464527	−194.368904	−194.347495
p median, n q05	−212.877473	−212.877473	−380.654962	−214.556763	−214.448495	−212.976377	−213.323667	−212.904257
p median, n q95	−235.887717	−235.887717	−234.563988	−235.609633	−235.748009	−235.859932	−235.873999	−235.885265

- Numerical integration converges toward the analytic log-likelihood as the grid grows, but convergence is slow and requires very fine grids.
- At coarse and moderate grid sizes, numerical integration overestimates the log-likelihood.

11.f. Evaluation Speed

All times are in seconds per call on the 215-observation residual vector.

Parameter set	Stabilized	Justin's closed-form	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p q05, n median	0.0003	0.0005	0.0086	0.0427	0.0883	0.4326	0.9102	4.5478
p median, n median	0.0004	0.0006	0.0075	0.0358	0.0730	0.3674	0.7613	3.8117
p q95, n median	0.0003	0.0005	0.0064	0.0308	0.0663	0.3258	0.6520	3.2415
p median, n q05	0.0003	0.0005	0.0076	0.0358	0.0737	0.3567	0.7479	3.6527
p median, n q95	0.0004	0.0006	0.0059	0.0284	0.0611	0.2900	0.5937	2.9724

- The stabilized implementation is modestly faster than the closed-form implementation for moderate-shape inputs (median speedup approximately 1.5 times), reflecting the avoidance of scalar `mpmath` fallback calls.
- Both analytic implementations are orders of magnitude faster than numerical integration. Relative to 5,000-grid integration, the median speedup of the stabilized implementation is approximately 957 times.

11.g. *Large-Shape Robustness*

The closed-form BEGE density contains a confluent hypergeometric term that is accurate for small and moderate shape states but becomes numerically fragile as the recursive shapes grow, because large log terms in the expression nearly cancel. In that regime, limited-precision evaluation can produce artificial pointwise log densities that are orders of magnitude from their true values.

The stabilized implementation applies the following guarded evaluation rule:

- retain the exact hypergeometric expression for small-shape observations;
- substitute the saddlepoint density when total shape enters the fragile range, using a smooth log-sum-exp transition rather than a hard cutoff;
- override exact values with the saddlepoint value whenever the two disagree by more than 2 log units in the guarded region;
- use the Gaussian density when total shape exceeds 500 and both standardized skewness and excess kurtosis are below 0.03.

11.g.i. *Saddlepoint Approximation:*

The saddlepoint approximation provides an analytical, grid-free approximation to a density derived from the cumulant-generating function (CGF). For a target observation u_t, the method finds the unique real number $\hat{s}$ — the saddlepoint — satisfying $K'(\hat{s}) = u_t$, where $K(s) = \log \mathbb{E}[e^{s u_t}]$ is the CGF of u_t.

This is a single nonlinear equation solved per observation by Newton's method. The log density is then given by the Lugannani–Rice formula:

$$\log f(u_t) \approx K(\hat{s}) - \hat{s}\, u_t - \frac{1}{2} \log\{\, 2\pi K''(\hat{s})\, \}. \tag{80}$$

The approximation is exact for Gaussian and gamma distributions and is highly accurate for sums of gamma shocks. For the BEGE shock, the CGF is available in closed form and its first two derivatives reduce to simple rational functions of s, so each saddlepoint evaluation costs a fixed small number of floating-point operations regardless of the shape values. This stands in contrast to convolution-based numerical integration, where accuracy grows slowly with grid size (compare the 100- and 50,000-point columns in the log-likelihood table above).

The saddlepoint approximation also respects the analytic normal limit of the BEGE distribution: as the gamma shapes grow while $\sigma_p^2 p_t + \sigma_n^2 n_t$ remains fixed, the standardized skewness and excess kurtosis shrink toward zero, and the saddlepoint density converges to the Gaussian density with the same conditional variance.

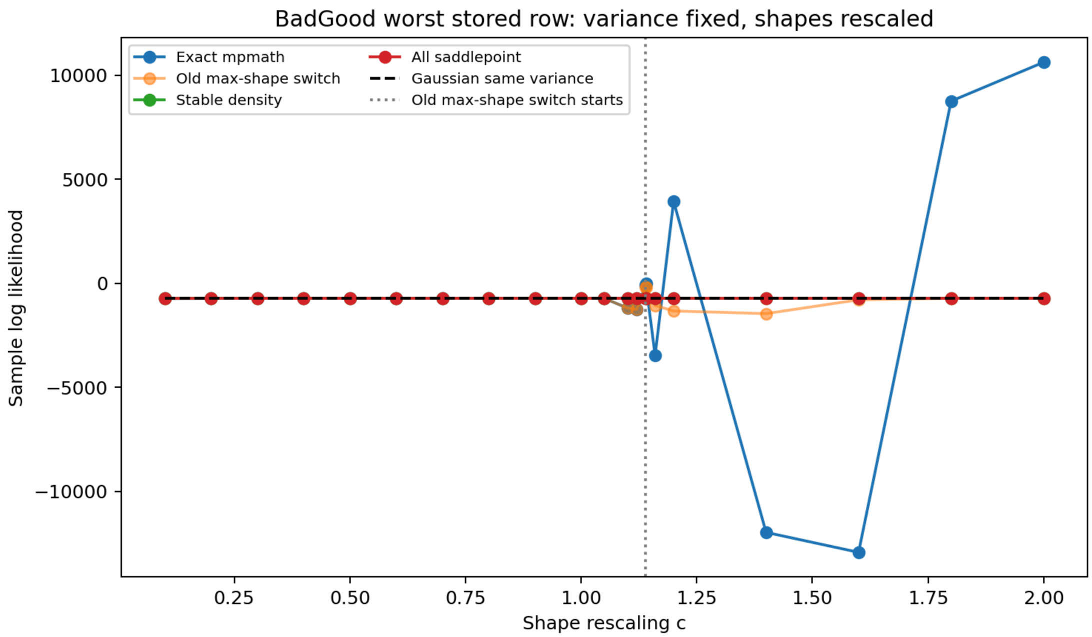
11.g.ii. *Saddlepoint Threshold Experiment:*

To verify the guarding rule, consider the variance-preserving rescaling

$$p_t(c) = c\, p_t, \quad n_t(c) = c\, n_t, \quad \sigma_p(c) = \sigma_p/\sqrt{c}, \quad \sigma_n(c) = \sigma_n/\sqrt{c}, \tag{81}$$

which holds the conditional variance path constant while scaling the gamma shapes. The BEGE distribution should converge monotonically to a Gaussian law as c increases. The experiment uses the seed 45, draw 25, ARX(2,1) path and varies c from 0.1 to 2.0. The five log-likelihood columns correspond to: (i) exact high-precision mpmath branches, (ii) Justin's closed-form implementation with the original single-variable cutoff ($\max(p_t, n_t) \geq 180$), (iii) the current stabilized implementation, (iv) the saddlepoint approximation, and (v) the Gaussian density with the same conditional variance.

c	Med. total shape	Max shape	Original cutoff obs.	Exact mpmath	Original cutoff	Stabilized	Saddlepoint	Gaussian
0.1000	19.4893	15.7964	0	−728.2549	−728.2549	−728.2549	−727.1415	−720.5045
0.4000	77.9574	63.1856	0	−724.0184	−724.0184	−724.0388	−723.9614	−720.5045
0.8000	155.9147	126.3712	0	−722.6555	−722.6555	−722.6530	−722.9801	−720.5045
1.0000	194.8934	157.9640	0	−722.4380	−722.4380	−722.4342	−722.7255	−720.5045
1.1000	214.3827	173.7604	0	−1187.7835	−1187.7835	−722.3558	−722.6246	−720.5045
1.1395	222.0811	180.0000	1	−94.4486	−174.4248	−722.3280	−722.5884	−720.5045
1.2000	233.8721	189.5568	162	3934.4808	−1329.2867	−722.2885	−722.5364	−720.5045
1.8000	350.8081	284.3352	197	8749.6077	−722.0183	−722.0077	−722.1699	−720.5045
2.0000	389.7868	315.9280	200	10616.6236	−721.9500	−721.9408	−722.0858	−720.5045



The table shows that the original single-variable cutoff is insufficient: exact-branch failures appear around $c = 1.10$ (total shape ≈ 214), before the $\max(p_t, n_t) = 180$ threshold is reached at $c = 1.1395$. The stored likelihood at $c = 1$ was 4920.7089, consistent with latent hypergeometric failures at the base parameter values. The implemented rule therefore conditions on total shape:

- guard exact hypergeometric values once $p_t + n_t \geq 40$;
- smooth-blend exact and saddlepoint values between total shape 50 and 80;
- replace exact values when exact and saddlepoint disagree by more than 2 log units in the guarded region;
- use the Gaussian limit only after total shape 500 and only when standardized skewness and excess kurtosis are both below 0.03.

11.h. Diagnostic Evidence

The table below examines BadGood results from earlier production runs. The stored log-likelihood is the value written by the closed-form implementation at estimation time. The same parameter vectors are then recomputed on the same effective-sample residuals using four alternative evaluators: the stabilized density, the all-saddlepoint density, the Gaussian density with the same conditional variance path, and exact high-precision mpmath branches.

Mean	Seed	Draw	Stored	Stabilized	Saddlepoint	Gaussian	Exact mpmath	Max shape	Med. total shape	Med. σ_t^2
ARX(2,1)	45	25	4920.7089	−722.4342	−722.7255	−720.5045	−722.4380	157.9640	194.8934	135.4737
Constant	29	31	4846.4315	−770.7119	−770.6706	−770.2032	−1889.5667	171.9426	234.7388	225.1110
ARX(2,1)	15	3	3098.3394	−599.0093	−598.9692	−598.1718	−1186.8621	199.7122	220.0504	42.6849
ARX(2,2)	35	31	2892.0304	−734.4489	−734.2511	−730.1723	−807.8124	173.4795	205.7171	146.5852

ARX(1,1)	20	14	1330.8412	−590.1301	−590.0846	−592.7498	−584.8732	170.2031	206.8959	40.1486
ARX(2,1)	41	2	1113.7517	−684.6683	−684.6022	−680.7040	−684.8619	146.8413	164.9777	90.3592
ARX(1,1)	13	32	846.3034	−694.3325	−694.2827	−692.9410	−791.0434	156.0623	190.5299	104.1080
Constant	1	9	651.3535	−561.1096	−561.0324	−561.6200	−1101.1363	209.9112	191.0321	30.1035

Across 8,000 BadGood rows in this diagnostic pass, the stored log-likelihoods contained 150 rows above −150, including 13 rows above zero. The stabilized recomputation produced zero rows above −150, with a maximum of −161.8929.

The table below reports the observations with the largest exact-versus-stabilized gaps. The Fourier-inversion column provides an additional pointwise check: it numerically inverts the BEGE characteristic function via quadrature over the frequency domain, and is independent of both the hypergeometric and convolution-based implementations.

Mean	Seed	Draw	Obs.	u_t	p_t	n_t	σ_t^2	Exact mpmath ℓ_t	Stabilized ℓ_t	Saddlepoint ℓ_t	Gaussian ℓ_t	Fourier ℓ_t
ARX(2,1)	15	3	177	0.0913	42.9229	187.7988	42.9373	90.3648	−2.8008	−2.8008	−2.7989	−2.8007
ARX(2,1)	15	3	171	0.2476	42.9393	189.9662	42.9916	87.2647	−2.8052	−2.8052	−2.8002	−2.8051
ARX(2,1)	15	3	160	0.1429	44.3305	199.7122	44.4494	77.4285	−2.8191	−2.8191	−2.8163	−2.8191
Constant	1	9	185	−0.5080	188.8323	17.0463	30.2154	32.6685	−2.6480	−2.6480	−2.6274	−2.6503
Constant	1	9	148	0.5209	180.3713	17.1093	30.2612	30.6176	−2.6069	−2.6069	−2.6284	−2.6090
Constant	1	9	85	1.0855	174.4991	17.0414	30.1040	28.9611	−2.5957	−2.5957	−2.6408	−2.5979

A conditional variance in the range 30–44 cannot support pointwise log densities between 29 and 90; such values are numerically impossible. The stabilized, saddlepoint, and Fourier-inversion evaluations all agree at approximately −2.80, confirming that the anomalies originate from a numerical failure of the hypergeometric evaluation rather than any feature of the BEGE dynamics.

11.i. Summary

- Convolution-based numerical integration converges to the analytic log-likelihood but is too slow for production estimation and remains sensitive to grid size; it serves as an independent convergence benchmark.
- The shape-tail consistency plot confirms agreement across a wide shape range for all implementations; the 5,000-grid numerical integration line is included for comparison.
- The saddlepoint threshold should be conditioned on total shape $p_t + n_t$ and on exact-versus-saddlepoint disagreement; a single-variable maximum-shape cap is an unreliable guard condition.
- Large-shape log-likelihood evaluation should not rely solely on the closed-form hypergeometric expression. The guarded saddlepoint and Gaussian-limit rules in the stabilized implementation prevent spurious large-likelihood values from influencing BEGE estimates.

12. Constant BEGE Best Model Summary

Generated: `2026-06-18T10:10:04` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate across mean processes. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

12.a. Selected Best Model

Best admissible estimate ranked by log likelihood across mean processes.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	30	30	−181.6884	381.3768	411.7126

Empirical path quantiles:

Series	5%	Median	95%
p_t	2.6646	2.6646	2.6646
n_t	0.2821	0.2821	0.2821
σ_t^2	0.3952	0.3952	0.3952
s_t^2	−0.2286	−0.2286	−0.2286
k_t^2	0.9308	0.9308	0.9308

Mean process:

$$\pi_{t+1} = 0.0041 + 0.3232\,\pi_t + 0.1633\,\pi_{t-1} + 0.4107\,SPF_t + 0.1942\,SPF_{t-1} + u_{t+1} \tag{82}$$

BEGE volatility process:

$$\begin{aligned} u_t &= 0.2712\,\omega_{p,t} - 0.8404\,\omega_{n,t}, \\ \omega_{p,t} &\sim \tilde{\Gamma}(\bar{p}, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(\bar{n}, 1), \\ \bar{p} &= 2.6646, \qquad \bar{n} = 0.2821, \\ \mathrm{Var}_t(u_t) &= (0.2712)^2\,2.6646 + (0.8404)^2\,0.2821. \end{aligned} \tag{83}$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.0041	0.1115
ρ_1	0.3232	0.1214

ρ_2	0.1633	0.0829
ϕ_1	0.4107	0.3014
ϕ_2	0.1942	0.2957
$\bar{p}$	2.6646	0.1839
$\bar{n}$	0.2821	0.1597
σ_p	0.2712	0.0218
σ_n	0.8404	0.3557

13. Symmetric BEGE Best Model Summary

Generated: `2026-06-18T10:10:15` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

13.a. Selected Best Model

Best admissible estimate ranked by stabilized log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	35	12	−167.9012	359.8025	400.2501

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.6347	1.1647	5.6629
n_t	0.3243	0.5949	2.8917
σ_t^2	0.1638	0.3005	1.4610
s_t^2	−0.2097	−0.0432	−0.0235
k_t^2	0.1876	0.3441	1.6727

Mean process:

$$\pi_{t+1} = 0.2006 + 0.1978\,\pi_t + 0.2405\,\pi_{t-1} + 0.2257\,SPF_t + 0.1197\,SPF_{t-1} + u_{t+1} \tag{84}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.3592\,\omega_{p,t} - 0.5026\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{85}$$

$$\begin{gathered} p_t = 0.1762 + 0.6852\,p_{t-1} + \frac{0.4128}{2(0.3592)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0809}{2(0.3592)^2}\left(u_{t-1}^{-}\right)^2, \\ n_t = 0.0900 + 0.6852\,n_{t-1} + \frac{0.4128}{2(0.5026)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0809}{2(0.5026)^2}\left(u_{t-1}^{-}\right)^2 \end{gathered} \tag{86}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.2006	0.0397
ρ_1	0.1978	0.0273
ρ_2	0.2405	0.0389
ϕ_1	0.2257	0.2154
ϕ_2	0.1197	0.1590
p_0	0.1762	0.0688
n_0	0.0900	0.0393
ρ	0.6852	0.0587
ϕ^+	0.4128	0.0560
ϕ^-	0.0809	0.0871
σ_p	0.3592	0.0670
σ_n	0.5026	0.0621

14. BadGood BEGE Best Model Summary

Generated: `2026-06-18T10:10:22` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

14.a. Selected Best Model

Best admissible estimate ranked by stabilized log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,1)	39	14	−163.5594	351.1188	391.5664

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.4097	0.5799	1.7285
n_t	0.2985	0.7108	3.8717
σ_t^2	0.1416	0.2554	1.0320
s_t^2	−0.2196	0.0384	0.0683
k_t^2	0.1754	0.2991	1.1456

Mean process:

$$\pi_{t+1} = 0.2119 + 0.2240\,\pi_t + 0.2742\,\pi_{t-1} + 0.2699\,SPF_t + u_{t+1} \tag{87}$$

BEGE volatility process:

$$\begin{aligned} u_t &= 0.4778\,\omega_{p,t} - 0.4042\,\omega_{n,t}, \\ \omega_{p,t} &\sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{aligned} \tag{88}$$

$$\begin{aligned} p_t &= 0.1575 + 0.5795\,p_{t-1} + \frac{0.2143}{2(0.4778)^2}\,u_{t-1}^2, \\ n_t &= 0.1900 + 0.2831\,n_{t-1} + \frac{0.6633}{2(0.4042)^2}\,u_{t-1}^2 \end{aligned} \tag{89}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.2119	0.0283
ρ_1	0.2240	0.0281
ρ_2	0.2742	0.0140
ϕ_1	0.2699	0.0447
p_0	0.1575	0.0329
n_0	0.1900	0.0469
ρ_p	0.5795	0.0474
ρ_n	0.2831	0.0315
ϕ_p	0.2143	0.0433
ϕ_n	0.6633	0.0058
σ_p	0.4778	0.0509
σ_n	0.4042	0.0328

15. Inflation/Deflation BEGE-GJR Best Model Summary

Generated: `2026-06-18T10:11:34` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

15.a. Selected Best Model

Best admissible estimate ranked by stabilized log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	3	39	−167.9018	361.8035	405.6218

Empirical path quantiles:

Series	5%	Median	95%
p_t	4.5793	7.2032	28.6664
n_t	0.0584	0.0697	0.6744
σ_t^2	0.1699	0.2766	1.0806
s_t^2	−1.2028	−0.0927	0.0660
k_t^2	0.3575	0.4378	3.8913

Mean process:

$$\pi_{t+1} = 0.1246 + 0.2617\,\pi_t + 0.1743\,\pi_{t-1} + 0.2915\,SPF_t + 0.1749\,SPF_{t-1} + u_{t+1} \tag{90}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.1486\,\omega_{p,t} - 0.9893\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{91}$$

$$\begin{aligned} p_t &= 1.2658 + 0.7076\,p_{t-1} + \frac{0.4250}{2(0.1486)^2}\left(u_{t-1}^{+}\right)^2, \\ n_t &= 0.0508 + 0.1317\,n_{t-1} + \frac{1.2626}{2(0.9893)^2}\left(u_{t-1}^{-}\right)^2 \end{aligned} \tag{92}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.1246	0.0726
ρ_1	0.2617	0.0767
ρ_2	0.1743	0.0758
ϕ_1	0.2915	0.3736
ϕ_2	0.1749	0.3151
p_0	1.2658	0.7514
n_0	0.0508	0.0486
ρ_p	0.7076	0.1141
ρ_n	0.1317	0.0583
ϕ_p^+	0.4250	0.1637
ϕ_n^-	1.2626	0.8901
σ_p	0.1486	0.0423
σ_n	0.9893	0.5903

16. Constant-p Full BEGE Best Model Summary

Generated: `2026-06-18T10:11:11` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

16.a. *Selected Best Model*

Best admissible estimate ranked by stabilized log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	28	25	−166.7909	357.5819	398.0296

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.4727	0.4727	0.4727
n_t	0.4590	0.9268	7.3165
σ_t^2	0.1990	0.2674	1.2023
s_t^2	−0.6797	0.0355	0.0878
k_t^2	0.2795	0.3396	1.1602

Mean process:

$$\pi_{t+1} = 0.1831 + 0.1660\,\pi_t + 0.0787\,\pi_{t-1} + 0.2999\,SPF_t + 0.3076\,SPF_{t-1} + u_{t+1} \tag{93}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.5281\,\omega_{p,t} - 0.3825\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{94}$$

$$\begin{aligned} p_t &= 0.4727, \\ n_t &= 0.2113 + 0.4747\,n_{t-1} + \frac{0.7577}{2(0.3825)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.2362}{2(0.3825)^2}\left(u_{t-1}^{-}\right)^2 \end{aligned} \tag{95}$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.1831	0.0290

ρ_1	0.1660	0.0376
ρ_2	0.0787	0.0273
ϕ_1	0.2999	0.1631
ϕ_2	0.3076	0.1028
p_0	0.4727	0.1424
n_0	0.2113	0.0517
ρ_n	0.4747	0.0291
ϕ_n^+	0.7577	0.0098
ϕ_n^-	0.2362	0.0348
σ_p	0.5281	0.1365
σ_n	0.3825	0.0364

17. Constant-n Full BEGE Best Model Summary

Generated: `2026-06-18T10:11:30` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

17.a. Selected Best Model

Best admissible estimate ranked by stabilized log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	31	40	−170.2704	364.5408	404.9885

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.6223	1.1422	5.6821
n_t	0.4055	0.4055	0.4055
σ_t^2	0.2157	0.2840	0.8798
s_t^2	−0.0950	−0.0455	0.3862
k_t^2	0.3302	0.3840	0.8532

Mean process:

$$\pi_{t+1} = 0.1460 + 0.1849\,\pi_t + 0.0812\,\pi_{t-1} + 0.9787\,SPF_t - 0.3699\,SPF_{t-1} + u_{t+1} \tag{96}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.3623\,\omega_{p,t} - 0.5750\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{97}$$

$$\begin{aligned} p_t &= 0.1443 + 0.7377\,p_{t-1} + \frac{0.4074}{2(0.3623)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0184}{2(0.3623)^2}\left(u_{t-1}^{-}\right)^2, \\ n_t &= 0.4055 \end{aligned} \tag{98}$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.1460	0.0336

ρ_1	0.1849	0.0432
ρ_2	0.0812	0.0439
ϕ_1	0.9787	0.1048
ϕ_2	−0.3699	0.0891
p_0	0.1443	0.0321
n_0	0.4055	0.0525
ρ_p	0.7377	0.0172
ϕ_p^+	0.4074	0.0285
ϕ_p^-	0.0184	0.0323
σ_p	0.3623	0.0408
σ_n	0.5750	0.0736

18. Full BEGE Best Model Summary

Generated: `2026-06-18T10:11:22` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `8000`

This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage and reported in the parameter table.

CSV outputs:

- constant cleaned rows
- ARX(1,1) cleaned rows
- ARX(2,1) cleaned rows
- ARX(2,2) cleaned rows

18.a. Selected Best Model

Best admissible estimate ranked by stabilized log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC
ARX(2,2)	13	33	−165.2805	360.5611	411.1206

Empirical path quantiles:

Series	5%	Median	95%
p_t	0.9806	1.1858	2.3678
n_t	0.0849	0.2556	2.8363
σ_t^2	0.1834	0.2969	1.1546
s_t^2	−0.9826	0.0427	0.1478
k_t^2	0.1925	0.3661	2.0843

Mean process:

$$\pi_{t+1} = 0.1937 + 0.1415\,\pi_t + 0.1380\,\pi_{t-1} + 0.2283\,SPF_t + 0.3635\,SPF_{t-1} + u_{t+1} \tag{99}$$

BEGE volatility process:

$$\begin{gathered} u_t = 0.3802\,\omega_{p,t} - 0.5787\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{100}$$

$$\begin{gathered} p_t = 0.0347 + 0.9561\,p_{t-1} + \frac{0.0000}{2(0.3802)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0334}{2(0.3802)^2}\left(u_{t-1}^{-}\right)^2, \\ n_t = 0.0295 + 0.5611\,n_{t-1} + \frac{0.7784}{2(0.5787)^2}\left(u_{t-1}^{+}\right)^2 + \frac{0.0720}{2(0.5787)^2}\left(u_{t-1}^{-}\right)^2 \end{gathered} \tag{101}$$

Parameter table:

Parameter	Estimate	Std. Error

c	0.1937	0.0783
ρ_1	0.1415	0.0611
ρ_2	0.1380	0.0618
ϕ_1	0.2283	0.2319
ϕ_2	0.3635	0.1626
p_0	0.0347	0.0277
n_0	0.0295	0.0258
ρ_p	0.9561	0.0242
ρ_n	0.5611	0.0879
ϕ_p^+	0.0000	NA
ϕ_p^-	0.0334	0.0159
ϕ_n^+	0.7784	0.1702
ϕ_n^-	0.0720	0.0947
σ_p	0.3802	0.0621
σ_n	0.5787	0.1108